Space Emanation Theory

Field capacity, source organization and gravitational observables

Erik Echeverria

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Abstract

Space Emanation Theory begins with a physical question: how do gravity and motion affect the processes recorded by clocks? I propose that gravitational burden, translation and local event progression share a limited field interaction capacity. The source law assigns a source quantity to a specified mass and volume; its field response determines the burden entering that capacity budget. This paper develops the connection from sources to physical readings and then to observations.

For an isolated spherical source, physical work-current balance and consistency across a common source family determine the exterior burden and radial response. The calibrated probe action gives conserved energy, escape speed and the perihelion equation, with a Mercury benchmark of approximately 42.98 arcseconds per century. A specified rotating extension gives the Kerr exterior, horizon source readings and shadow sizes compatible with the cited Galactic-center inference.

For collective baryonic sources, an effective-area prescription relates source organization to boundary burden. Twenty single-radius galaxy examples have a median absolute central speed discrepancy of 4.79%; among nineteen matched cases, the SET and Newtonian baryonic medians are 5.13% and 40.67%. A homogeneous cosmological source-clock prescription then calculates relative Type Ia supernova brightness. With the same fixed inputs across two compilations, all fifteen grouped discrepancies decrease; the catalogue medians fall from 38.53% to 2.09% and from 36.39% to 2.96%. The comparison baselines, source prescriptions, measured inputs and uncertainty treatments are given with each calculation. These applications investigate whether effects attributed to dark matter and dark energy can arise through the collective response of the field to baryonic sources.

Reading guide

1 Why field interaction capacity

2 Capacity clocks and source quantities

3 Physical work and the spherical response

4 Clock normalization, propagation and the orbital gradient

5 Perihelion from the spherical field

6 Strong fields and rotating black hole exteriors

7 Effective area and collective source organization

8 Galaxy calculations and observed velocities

9 Collective fields and energy accounting

10 Cosmological source evolution and physical clocks

11 Supernova brightness and observations

12 Physical interpretation and results

Appendices A–J retain source-family controls, alternative orbit calculations, perihelion clock accumulation, free-fall coordinates, galaxy inputs, collective energy accounting, a finite flat-speed illustration, gradient normalization, numerical checks and cosmological distance checks.

1 Why field interaction capacity

I began SET with a physical question: what happens to the processes of a clock when it records fewer ticks per reference second? A clock has to do something to produce each tick. The observations used to compare clocks also require physical interactions. I wanted to follow those processes through their interaction with the field and ask how the metric records the result.

“There are no motionless events” expresses my starting intuition: an observed event involves a physical occurrence or change. A body can be at rest as a whole while its internal processes continue. Observing that body also requires an interaction. Here, physical evolution includes interactions and recorded changes; it does not require a hidden classical trajectory within every quantum state. A stationary energy eigenstate need not produce distinguishable ticks, and a global phase change alone need not be observable. The hypothesis concerns the processes through which matter realizes events and observations.

My proposal is that the field has a limited interaction capacity shared by gravitational burden, translation and local event progression. A held clock has less capacity available for event progression where gravitational burden is greater. A moving probe shares the remaining capacity between translation and its local processes. These comparisons use a specified reference time.

A massive probe can approach the local speed c, while the adopted relativistic energy law makes the energy required to reach it unbounded. I interpret the accompanying reduction in event progression per reference second as increased participation of translation in the shared capacity budget.

In this interpretation, proper time records the progression of those physical processes. Fewer calibrated events per reference second means less accumulated proper time. I am not adding a separate time process that first slows and then acts on matter. Clocks following different paths can accumulate different event counts between the same endpoint events.

The observer’s clock participates in the field it is used to measure. This matters because the instrument and the phenomenon are subject to the same local conditions. In the stationary construction developed below, light’s propagation expressed as local distance per reference second and the held clock’s progression share the same factor. That factor cancels when propagation is timed by the local clock. Comparisons between clocks at different locations, or along different paths, and signal travel times referred to a specified clock reveal the gravitational effect. Section 4.4 distinguishes this local measurement from the radial coordinate speed.

Why introduce field interaction capacity when Newtonian gravity and general relativity already provide equations of motion?

Newtonian gravity supplies the successful weak-field mechanics used in many gravitational calculations. It can be expressed through a potential field, and it already accounts for the spatial distribution of extended sources, including flattened disks. Its limitation for the present purpose is that it does not itself supply the relativistic relations among clock progression, light propagation and motion. Those relations require a more complete description.

General relativity supplies that description through a dynamical metric coupled to energy and momentum. Its relativistic effects cannot be reduced to a delay added to Newton’s force law: even a stationary gravitational field changes the relations among measured distance, clock readings and orbital motion. SET retains these connections in the spherical construction presented here.

My incentive for introducing capacity is to investigate a physical account of those relations. I ask whether the metric can be understood through the participation of matter, clocks and signals in a limited field interaction capacity. The proposed order of explanation begins with sources and their field response, follows how that response affects physical processes, and expresses the resulting clock and propagation relations geometrically.

Curvature therefore remains part of the description. With the source equation, propagation coupling and action specified in this paper, the capacity formulation recovers the Schwarzschild exterior and its test-

particle dynamics. That recovery establishes the connection for the spherical problem considered here. It also identifies which physical relations the capacity interpretation must reproduce.

I denote gravitational burden by S, expressed in speed units. Its squared magnitude enters the capacity budget. Through the adopted energy coupling, it determines the escape-energy scale; its spatial variation determines the weak-field acceleration. The familiar spherical expression $S = \sqrt{(2GM/r)}$ is one solution for this quantity. The burden is established by the sources and their response, and the probe's escape speed and orbital motion follow through the stated dynamical relations.

This viewpoint leads to the collective question that motivates the second part of the paper. Can the source–boundary relation used for an individual body also apply to a system of disjoint masses, with their organization entering through an effective boundary area?

At Earth's approximately spherical surface, the spherical source normalization and surface area give the boundary burden. For a collective system, I apply SET's proposed source–boundary relation to the organization of its baryonic sources. At fixed baryonic mass and reference radius, the reference normalization is retained and the boundary burden follows from $Q_{ref} = A_{eff} S_{eff}$. A smaller effective area therefore gives a larger boundary burden without changing Q_{ref} or introducing a separate enhancement factor. This is why source organization is a natural route to investigate within SET: the proposed change in gravitational response enters through the relation between source normalization and boundary area, without adding unseen gravitating matter. The familiar spherical expression $S = \sqrt{(2GM/R)}$ is the spherical case of that reading; it is not held fixed while the effective area changes.

An effective boundary must describe the physical response. Rearranging the sources can change their collective field; merely redrawing a surface around an unchanged system cannot change that field at a fixed location. The area used in the calculation must consequently be connected to the source arrangement. In the galaxy examples, I estimate it with an oblate surface constrained by imaging and inclination.

The galaxy calculation examines whether the proposed capacity and boundary relation allows the adopted baryonic inventory to account for the measured orbital speed. Where matching models are available, I compare its result with spatially resolved Newtonian gas, disk and bulge contributions at the same radius.

The same question extends to the light received from distant sources. In galaxies, orbital motion probes a spatial response. In the supernova calculation, brightness probes a history through source-driven motion, physical clocks and propagation. I want to know whether the source–capacity relationship can account for both kinds of observation without assigning them to separate invisible components. The calculations below make that question concrete in each setting.

I begin with the source law and invariant, derive the spherical response and orbital motion, then examine strong fields and rotation. I next develop the effective-area calculation and its galaxy comparisons. The final application follows a collective source history into the observed supernova brightness trend. Detailed controls, alternative representations and numerical checks are retained in the appendices so that the main argument can be followed without losing the steps needed to reproduce it.

2 Capacity clocks and source quantities

2.1 The invariant and recorded events

For the isolated spherical problem, take a static reference time T normalized at infinity. A probe records proper time τ through its calibrated event accumulation. Let dℓ be the displacement norm in the static spatial description, and define β = dτ/dT and Vrel = dℓ/dT. A nearby held observer uses a different clock rate; its measured speed is derived below.

$$S^2 + V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2 . \quad (1)$$

Each term has units m^2/s^2. S is the local gravitational burden expressed in speed units; S^2 is its contribution to the capacity budget. In SET's interpretation, sources establish the burden, and the energy coupling determines the escape speed associated with it, as derived in Section 4.4. Translation contributes $Vrel^2$, while $c^2\beta^2$ represents event progression. These are scalar contributions to a shared budget.

The sources and their spatial response determine S. The spherical expression derived below is one solution for that field. Section 9.1 shows the source contribution inside the invariant itself.

Place the source burden on the right. The left then shows how the probe shares the remaining capacity between translation and event progression:

$$V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2 - S^2 . \quad (2)$$

$$u = \frac{S^2}{c^2}, \quad \alpha^2 = 1 - u, \quad \beta^2 = \alpha^2 - \frac{V_{\mathrm{rel}}^2}{c^2} . \quad (3)$$

For a held probe, Vrel = 0 and β = α. Thus α is the held-clock rate relative to T. A local static observer uses dtlocal = α dT, so the local translation speed is vlocal = dℓ/dtlocal = Vrel/α. Substitution gives:

$$\beta^2 = \alpha^2\left(1 - \frac{v_{\mathrm{local}}^2}{c^2}\right), \quad \beta = \frac{\alpha}{\gamma_{\mathrm{local}}} . \quad (4)$$

where $\gamma local = 1/\sqrt{(1 - vlocal^2/c^2)}$. Neither expression permits a massive probe with a real, positive proper-time increment to have vlocal ≥ c.

In SET, the field has a physical interaction limit. It does not need to classify an interaction as gravity, translation or event progression. These terms describe how the available capacity is shared. Different combinations of gravitational burden and translation can leave the same capacity for local events, compared using the same reference time. The resulting event-rate factor can therefore be the same even when the surrounding gravitational fields differ. Equal remaining capacity does not imply identical physical circumstances.

To give event progression an operational meaning, let N count the cycles of an ideal clock and let ν_0 be its calibrated number of cycles per unit local clock reading. Then $d\tau = dN/\nu_0$ and $dN = \nu_0 d\tau$. The reading records the events, so:

$$\frac{dN}{dT} = \nu_0\beta . \quad (5)$$

The clock completes $\nu_0\beta$ events per reference second. I take the common factor β for ideal clocks as part of the universal coupling to capacity. Real clock comparisons must also control temperature, acceleration-induced defects and other calibration changes. The scale ν_0 belongs to the clock: c alone specifies no universal tick count. At u = 0 and Vrel = 0, β = 1 and this calibrated process has its maximal rate relative to the unloaded reference.

For example, take u = 0.36 and vlocal = 0.6c. Then α = 0.8, Vrel = 0.48c and β = 0.64. Burden, translation and event progression occupy 0.36, 0.2304 and 0.4096 of c^2, respectively. Their sum is one.

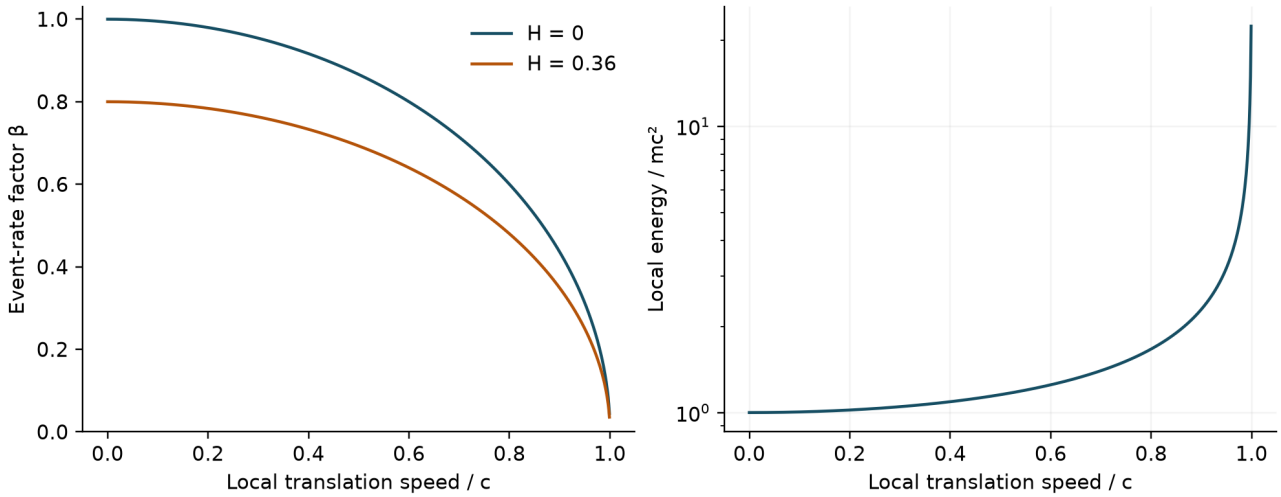


Figure 1. Left: event-rate factor β as local translation increases, for two fixed occupancies. Right: the local massive-probe energy obtained from the action in Section 3.6. The divergence is a limit as vlocal approaches c from below. In this retained plot, H denotes the occupancy u used in the text.

2.2 Displacement and proper velocity

Multiplying equation (2) by dT^2 turns the translation rate into a displacement:

$$d\ell^2 + c^2 d\tau^2 = (c^2 - S^2)dT^2 . \quad (6)$$

$$c^2 d\tau^2 = \alpha^2 c^2 dT^2 - d\ell^2 . \quad (7)$$

This gives the time–motion partition of the interval. To calculate a path, write $d\ell^2 = q_{ij}\, dx^i dx^j$, where q_{ij} is a positive spatial quadratic form in the static region and repeated spatial indices are summed. Section 3 determines its spherical radial response through the source and physical-current construction.

Now introduce proper velocity $u_p{}^i = dx^i/d\tau$ and its norm $u_p{}^2 = q_{ij}\, u_p{}^i u_p{}^j$. Since $dx^i/dT = \beta u_p{}^i$, $V_{rel}^2 = \beta^2 u_p{}^2$. Equation (2) becomes:

$$\beta^2(c^2 + u_p^2) = c^2\alpha^2 . \quad (8)$$

$$\beta = \frac{\alpha}{\sqrt{1 + u_p^2/c^2}}, \quad u_p = \gamma_{local} v_{local} . \quad (9)$$

Here u_p measures displacement per unit of the moving probe's own event accumulation. The local static observer measures vlocal using its clock. Equation (9) relates the two rates; the local causal limit applies to vlocal.

Equations (2)–(9) follow by rearranging the invariant with these definitions. To proceed, I specify how mass establishes S, how burden enters the spatial response, and which action determines a free path.

2.3 The source law and its domain

Begin with a specified source domain Ω containing mass M in volume VΩ. I use the SET source law:

$$Q_\Omega^2 = 24\pi G M V_\Omega, \quad \mu_Q = \frac{Q_\Omega^2}{24\pi V_\Omega} = GM. \quad (10)$$

Here QΩ has units m³/s and μQ has units m³/s². With a spherical source-domain radius RΩ, VΩ = 4πRΩ³/3 and:

$$Q_\Omega = 4\pi\sqrt{2GMR_\Omega^3}. \quad (11)$$

The density form follows by substituting VΩ = M/ρ̄:

$$Q_\Omega = M\sqrt{\frac{24\pi G}{\bar{\rho}}}. \quad (12)$$

Keep the source domain attached to QΩ. For several sources, the mass-normalized sum is $\Sigma Q_a^2/V_a = 24\pi G\Sigma M_a$. This remains unchanged if we refine a numerical description of the same mass. For example, divide a uniform domain into N equal cells. Then Qcell = QΩ/N and $\Sigma Q_{cell}^2 = Q\Omega^2/N$, while $\Sigma Q_{cell}^2/V_{cell} = Q\Omega^2/V\Omega$. It is the normalized quantity, rather than bare Q², that survives arbitrary subdivision.

For a fixed stationary source, μQ = GM is the constant strength entering the field equation. The spherical construction below determines how the burden varies outside that source.

For an evaluation sphere at r, define a separate area-scaled readout:

$$Q_0(r) = 4\pi r^2 S(r). \quad (13)$$

QΩ describes the source domain; $Q_0(r)$ describes an evaluation screen. Both have units m³/s, but they answer different questions. To determine whether Q_0 stays fixed between screens, we must first solve the field.

A source-domain quantity, a screen reading and a receiving-clock rate refer to different physical operations. I keep their labels visible throughout the paper. In the compact-body calculation, M is the total source mass-energy parameter. The galaxy and cosmological applications instead use their explicitly adopted baryonic inventories.

Table 1. Source quantities and their readings

Reading	Definition and use
Source domain	$Q_\Omega^2 = 24\pi G M V_\Omega$ for the stated source volume.
Spherical exterior screen	$Q_0(r) = 4\pi r^2 S(r)$, evaluated after the field is solved.
Collective boundary	$Q_{\mathrm{ref}} = A_{\mathrm{eff}} S_{\mathrm{eff}}$ at the adopted mass and reference radius.
Held receiving clock	$Q_\star = Q_0/\alpha$, with the same spatial reading expressed per held-clock second.
Horizon screen	Q_H is the limiting source reading at the specified outer cap.
Cosmological material domain	$Q_m^2 = 24\pi G M_m V_m$; the retained evolution law is $dV_m/dT = Q_m$.

For the isolated exterior, T is normalized by the unloaded limit at infinity. In the homogeneous cosmological realization, T is the reference time of the source law and the comoving physical clock obeys dt = α dT. The

local clock conversion has the same form; the reference construction and the physical source domain are stated separately for the two applications.

2.4 The radial motion reading

Here S is gravitational burden in speed units, T is the stationary reference time, and τ records the probe's clock progression. For a held observer, define

$$A = \alpha^2 = 1 - \frac{S^2}{c^2}, \quad d\tau_{\mathsf{h}} = \alpha\, dT. \quad (14)$$

Use areal radius r, so a spherical screen has area $A_\Sigma = 4\pi r^2$. The dimensionless A and the area A_Σ are distinct. In the equatorial plane, write

$$V_{\mathsf{rel}}^2 = \zeta^2 \dot{r}^2 + r^2 \dot{\varphi}^2, \quad d\ell = \zeta\, |dr|. \quad (15)$$

Dots denote d/dT. The positive factor ζ converts an areal-radius displacement to the radial distance entering the local motion measurement. It will be determined by the field construction rather than fixed at the outset.

Notation will distinguish the occupancy $u = S^2/c^2$ from the spherical response function $\mathscr{H}(A)$, with $A = \alpha^2$. The symbol H is reserved for the cosmological rate. The spherical work-current is written $\mathscr{J}$, while J denotes source angular momentum in the rotating section. Proper velocity in the general static formulas is $u_{\boldsymbol{p}}$; the galaxy amplitude diagnostic is $b_{\boldsymbol{s}}$. The response tensor C in the collective field equation and the catalogue covariance matrix in the supernova statistic have separately specified roles.

3 Physical work and the spherical response

SET takes continuous sourcing of field burden as a physical principle. Here I make its stationary spherical application explicit: the source-surface relation fixes the boundary reading, the physical work-current carries the source charge through complete vacuum spheres, and the same capacity-dependent response applies across the specified source family. The field is solved before an orbit is calculated.

The calculation concerns an isolated, nonrotating, positive spherical source and a test body in its stationary vacuum exterior. The reference time is normalized at infinity, and the held-clock capacity is positive throughout the region considered. The source interior, other bodies and radiation reaction are outside this calculation.

3.1 Source surface and universal response

Let the source have mass parameter M and physical surface radius R. The source law uses the spherical reference volume and the surface burden:

$$V_{\mathsf{ref}} = \frac{4\pi R^3}{3}, \quad Q_\Omega^2 = 24\pi G M V_{\mathsf{ref}} = 32\pi^2 G M R^3. \quad (16)$$

$$Q_\Omega = 4\pi R^2 S(R). \quad (17)$$

Q_Ω is assigned to this source domain. M denotes the total source mass-energy parameter used consistently in the boundary and exterior equations. The reference volume is the volume specified by the source law; it is not replaced by an integral over an as-yet unknown physical radial measure.

For the positive source branch, equation (16) is $Q_\Omega = \sqrt{32\pi^2 G M R^3} = 4\pi\sqrt{2GMR^3}$. Its squared form and the ratio $Q_\Omega^2/(24\pi V_{\mathsf{ref}}) = GM$ are algebraic expressions of the same law. The surface relation places

that source law in the capacity budget through $S(R) = Q_\Omega/(4\pi R^2)$. A change to receiving-clock units, treated in Section 4.1, is a separate conversion of the readout.

Squaring equation (17) and comparing with equation (16) gives

$$16\pi^2 R^4 S(R)^2 = 32\pi^2 GMR^3, \quad S(R)^2 = \frac{2GM}{R}. \quad (18)$$

$$A_R = 1 - \frac{R_s}{R}, \quad R_s = \frac{2GM}{c^2}, \quad R > R_s. \quad (19)$$

This fixes the burden at an actual source surface. We next determine how the same source establishes the exterior response. Parameterize the radial factor by

$$\zeta^2 = \frac{\mathscr{H}(A)}{A}, \quad \mathscr{H}(A) > 0, \quad \mathscr{H}(1) = 1. \quad (20)$$

We take $\mathscr{H}$ to be positive, continuous and universal across the specified stationary spherical source family, with the differentiability required for the motion equations. Its only independent argument in this class is the local capacity A. The class excludes separate dependence on source size, density gradients, history or an additional field. This restriction is part of the proposed spherical response law; spherical symmetry alone does not impose it.

The unloaded limit has $A \to 1$ and $\zeta \to 1$ at infinity. The allowed source family is assumed to span a continuous interval of surface capacities extending to this limit. Section 3.4 uses that common family to determine $\mathscr{H}$; it does not move the boundary of one fixed source through vacuum.

3.2 Motion from calibrated event progression

The capacity relation assigns a clock-progression factor β to each proposed probe path. For a stable clock, calibrated event accumulation along that path is proportional to $\int \beta\, dT$. We use this accumulation to define the action of a structureless test body with constant rest mass and no additional direct field coupling.

The motion principle is stationarity under path variations with fixed endpoint events. With a positive constant κ carrying energy units, write

$$I_p = -\kappa \int \beta\, dT, \quad \delta I_p = 0. \quad (21)$$

Each interval contributes the same action per unit of calibrated event progression. This is the minimal coupling adopted here. The invariant determines β; stationarity determines which path the free probe follows. It does not require every physical path to maximize an event count.

Calibrate the coefficient with the probe's measured inertial mass. In the unloaded limit $S = 0$ and $\zeta = 1$, write $V_{\mathsf{rel}} = v$ and expand at low speed:

$$L_p = -\kappa\sqrt{1 - \frac{v^2}{c^2}} = -\kappa + \frac{\kappa v^2}{2c^2} + O\left(\frac{\kappa v^4}{c^4}\right). \quad (22)$$

The kinetic term must be $m_p v^2/2$. Therefore

$$\frac{\kappa}{c^2} = m_p, \quad \kappa = m_p c^2 . \quad (23)$$

The calibrated probe action is consequently

$$I_p = -\, m_p c^2 \int \beta \, dT, \quad L_p = -\, m_p c^2 \beta . \quad (24)$$

This fixes the coefficient without using an orbital precession. It also gives the gravitational coupling directly. For $S^2/c^2 \ll 1$ and $V_{\mathsf{rel}}^2/c^2 \ll 1$, expand the same action:

$$L_p = -\, m_p c^2 + \frac{m_p}{2}(V_{\mathsf{rel}}^2 + S^2) + O\left(\frac{m_p (S^2 + V_{\mathsf{rel}}^2)^2}{c^2} \right). \quad (25)$$

At leading order in the unloaded spatial measure, the specific potential and the acceleration are

$$\Phi = -\frac{S^2}{2}, \quad \boldsymbol{a} = -\nabla \Phi = \frac{1}{2} \nabla S^2 . \quad (26)$$

The attractive response follows because the spherical burden decreases outward. The higher-order orbital terms require the full action and the radial response, which we now determine together with the exterior field.

The same action can be expressed in phase cycles. Define $\nu m = mc^2/h$ and $dNm = \nu m \, d\tau$, with h Planck's constant. Then $-h\int dNm = -h(mc^2/h)\int d\tau = -mc^2 \int d\tau$. With $\varphi m = 2\pi Nm$ and $\hbar = h/(2\pi)$, it is also $-\hbar \int d\varphi m$.

The connection to capacity is $dNm/dT = \nu m \beta$: the same β controls this phase accumulation relative to T. Nm is a phase normalization; N in Section 2.1 counts recorded clock cycles. Observing phase requires a physical comparison sensitive to phase differences. This rewriting supplies neither an internal energy spectrum nor a quantization rule, and it assigns no observable tick train at νm to every particle.

3.3 Physical work and the exterior current

Combining equations (1) and (15) gives $\beta^2 = A - (\zeta^2 \dot{r}^2 + r^2 \dot{\varphi}^2)/c^2$. Differentiation of the action yields

$$p_r = \frac{m_p \zeta^2 \dot{r}}{\beta}, \quad p_\varphi = \frac{m_p r^2 \dot{\varphi}}{\beta} . \quad (27)$$

$$E = p_r \dot{r} + p_\varphi \dot{\varphi} - L_p = \frac{m_p c^2 A}{\beta} . \quad (28)$$

The final equality uses $\zeta^2 \dot{r}^2 + r^2 \dot{\varphi}^2 = c^2 (A - \beta^2)$. Locally, the motion factor is $\gamma = \alpha/\beta$, so $E = \alpha m_p c^2 \gamma$. A held probe therefore has $E_{\mathsf{hold}} = m_p c^2 \alpha$.

For a quasistatic outward lift, local work $F_{\mathsf{hold}} d\ell$ contributes $\alpha F_{\mathsf{hold}} d\ell$ to the reference energy. Equating this to dE_{hold} gives

$$\alpha F_{\mathsf{hold}} \zeta \, dr = m_p c^2 \frac{d\alpha}{dr} dr . \quad (29)$$

$$a_{\mathrm{hold}} = \frac{c^2}{\zeta}\frac{d\ln\alpha}{dr}, \quad \mathcal{J} \equiv \alpha a_{\mathrm{hold}} = \frac{c^2}{2\sqrt{\mathcal{H}(A)}}\frac{dA}{dr}. \quad (30)$$

$\mathcal{J}$ is the outward redshifted support acceleration. Defining the radial increment $d\chi = \alpha\, d\ell = \sqrt{\mathcal{H}}\, dr$ for outward displacements also gives $\mathcal{J} = -\frac{1}{2} dS^2/d\chi$. The quantity has units of acceleration; it is not the volume-per-time readout Q.

Sections 4.2 and 4.3 derive the finite and differential forms of this same current from the capacity relation and the held-clock normalization. They keep the distance χ explicit until the radial response has been selected.

We retain the spherical physical-current law of the earlier SET construction [10, 1]: every complete vacuum screen carries the same source charge in this work-current measure,

$$\oint_{\Sigma_r} \mathcal{J} \cdot d\boldsymbol{A}_\Sigma = 4\pi GM, \quad r^2\mathcal{J} = GM. \quad (31)$$

Equation (31) states the spherical field-response principle: the work-current carries the source charge through the vacuum exterior. Its value is fixed by the source law through $GM = Q_\Omega^2/(24\pi V_{\mathrm{ref}})$. This identifies the physical gradient to which conservation applies. Together with the universal response class, it defines the field problem whose solution follows next.

3.4 Consistency across the source family

Combine the work-current expression with the complete-screen law:

$$\frac{r^2c^2}{2\sqrt{\mathcal{H}(A)}}\frac{dA}{dr} = GM. \quad (32)$$

Separate variables and integrate from the actual surface R to the unloaded exterior limit:

$$\frac{dA}{\sqrt{\mathcal{H}(A)}} = \frac{R_s}{r^2}\, dr, \quad (33)$$

$$\int_{A_R}^{1} \frac{du}{\sqrt{\mathcal{H}(u)}} = R_s \int_R^\infty \frac{dr}{r^2} = \frac{R_s}{R}. \quad (34)$$

The independently retained source-surface relation gives $R_s/R = 1 - A_R$. Thus agreement of the source and field requires

$$\int_{A_R}^{1} \frac{du}{\sqrt{\mathcal{H}(u)}} = 1 - A_R. \quad (35)$$

Require this identity for the continuous family of source surfaces, using the same function $\mathcal{H}$. Differentiating with respect to the family's boundary capacity gives

$$-\frac{1}{\sqrt{\mathcal{H}(A_R)}} = -1, \quad \mathcal{H}(A_R) = 1. \quad (36)$$

The fundamental theorem of calculus applies because $\mathscr{H}$ is positive and continuous. If the admissible family covers $A_{\min} \le A_R < 1$, it determines the response throughout that interval:

$$\mathscr{H}(A) = 1, \ \zeta^2 = \frac{1}{A}. \quad (37)$$

Each member of the family has its own actual source radius, reference volume and source readout. The differentiation compares those physical configurations. It does not equate a fixed Q_Ω to every exterior sphere's area times its local burden.

The conclusion applies to the exterior capacities covered by the family. Extending it below $A_{\min}$ would require a corresponding extension of the admissible source family. This argument does not establish the stability of arbitrarily compact sources. Appendix A shows explicitly why agreement at only one source surface would not determine the entire function.

3.5 The exterior burden and screen reading

With $\mathscr{H} = 1$, the field equation has become

$$\frac{dA}{dr} = \frac{2GM}{c^2 r^2}, \ \ A(\infty) = 1. \quad (38)$$

Integrating from r to infinity gives the local profile:

$$1 - A(r) = \int_r^\infty \frac{2GM}{c^2 u^2}\, du = \frac{2GM}{rc^2}, \quad (39)$$

$$A(r) = 1 - \frac{2GM}{rc^2}, \ \ S(r)^2 = \frac{2GM}{r}. \quad (40)$$

The same calculation supplies the inverse-area work-current and the burden-gradient relation:

$$\mathscr{J} = \frac{GM}{r^2} = -\frac{1}{2}\frac{dS^2}{dr}, \quad -\frac{r^2}{2}\frac{dS^2}{dr} = GM. \quad (41)$$

Before selection, the physical gradient contained the factor $1/\sqrt{\mathscr{H}}$. Both that radial response and the profile are now fixed. The older carrier relation follows as an algebraic consequence:

$$S^4 = \frac{4G^2M^2}{r^2}, \ \ \mathscr{J} = \frac{S^4}{4GM}. \quad (42)$$

Define the local spherical screen readout, distinguishing it from the fixed source-domain quantity:

$$Q_0(r) = 4\pi r^2 S(r) = 4\pi\sqrt{2GMr^3}. \quad (43)$$

$$\frac{Q_0(r)^2}{24\pi V_{\mathrm{ref}}(r)} = \frac{rS(r)^2}{2} = GM, \ \ V_{\mathrm{ref}}(r) = \frac{4\pi r^3}{3}. \quad (44)$$

The source charge read this way is independent of the exterior screen. This constancy is a result of the solved field. $Q_0(r)$ itself grows as $r^{3/2}$ and is not a conserved volume discharge across successive spheres. At the material surface, $Q_0(R) = Q_\Omega$.

In terms of the screen quantity itself, the selected gradient equation can also be written as follows. This is the source-current relation expressed through Q_0 rather than a second field equation.

$$-\frac{r^2}{2}\frac{d}{dr}\left[\frac{Q_0^2}{(4\pi r^2)^2}\right]=\frac{Q_\Omega^2}{24\pi V_\Omega}. \quad (45)$$

For Earth, M = 5.9722×10²⁴ kg, R = 6.371×10⁶ m and G = 6.67430×10⁻¹¹ m³ kg⁻¹ s⁻² give QΩ ≈ 5.70567×10¹⁸ m³/s, S ≈ 11.19 km/s and u ≈ 1.39226×10⁻⁹. The weak-field gradient below gives 9.8203 m/s² at that mean spherical radius. These are normalization checks; Earth's rotation and latitude-dependent surface acceleration are outside this spherical example.

The surface of a body marks a change in the source distribution, not the end of its gravitational burden. Outside Earth, the burden continues according to the field law. Nothing special happens merely because we draw another enclosing surface ten kilometres higher. The field responds to the actual distribution of matter and its physical evolution; it does not respond to the boundary we choose for a calculation. That boundary tells us where we are reading the field.

3.6 The general static motion equation

The spherical calculation uses one radial response. The same action can be differentiated before specializing the static spatial form q_{ij}. In the following formulas m denotes the probe rest mass $m_{\boldsymbol{p}}$, $v^i = dx^i/dT$, and $u_{\boldsymbol{p}}{}^{\boldsymbol{i}} = dx^i/d\tau$.

Set $v^i = dx^i/dT$ and write the Lagrangian using the invariant:

$$L=-mc^2\sqrt{1-u-\frac{q_{ij}v^iv^j}{c^2}}. \quad (46)$$

The required derivatives are:

$$\frac{\partial L}{\partial v^i}=m\,q_{ij}u_{\mathrm{p}}^j. \quad (47)$$

$$\frac{\partial L}{\partial x^i}=\frac{mc^2}{2\beta}\partial_i u+\frac{m}{2\beta}(\partial_i q_{jk})v^jv^k. \quad (48)$$

Apply $d(\partial L/\partial v^i)/dT - \partial L/\partial x^i = 0$. Since $d/dT = \beta d/d\tau$ and $v^i = \beta u_{\boldsymbol{p}}{}^{\boldsymbol{i}}$, division by mβ gives:

$$\frac{d(q_{ij}u_{\mathrm{p}}^j)}{d\tau}-\frac{1}{2}(\partial_i q_{jk})u_{\mathrm{p}}^ju_{\mathrm{p}}^k=\frac{c^2}{2\beta^2}\partial_i u. \quad (49)$$

Raising the index groups the spatial-metric derivatives into its connection. With $Du_{\boldsymbol{p}}{}^{\boldsymbol{i}}/d\tau = du_{\boldsymbol{p}}{}^{\boldsymbol{i}}/d\tau + \Gamma^i{}_{j\boldsymbol{k}}u_{\boldsymbol{p}}{}^j u_{\boldsymbol{p}}{}^k$ and $D^iu = q^{ij}\partial_j u$:

$$\beta^2\frac{Du_{\mathrm{p}}^i}{d\tau}=\frac{c^2}{2}D^iu=\frac{1}{2}D^iS^2. \quad (50)$$

Using equation (8) and $D^i\ln\alpha = -D^iu/(2\alpha^2)$ gives the equivalent form:

$$\frac{Du_{\mathrm{p}}^i}{d\tau}=-(c^2+u_{\mathrm{p}}^2)D^i\ln\alpha. \quad (51)$$

This is the static motion law used in SET. The source contributes the burden gradient, and β accounts for the probe's event progression relative to T. The probe mass cancels. The derivation uses the invariant together with the spatial response and the stationary-action principle.

$Du_p^i/d\tau$ is a spatial covariant acceleration. It is not identical to coordinate acceleration d^2x^i/dT^2 or to the acceleration measured by an accelerometer. A freely falling probe has zero spacetime proper acceleration. The equation describes its trajectory relative to the static spatial readout. Time-dependent, rotating and drifting fields require the corresponding additional terms.

3.7 Energy from the same action

For a stationary field, the canonical energy is $E = v^i(\partial L/\partial v^i) - L$. Substituting equations (46)–(47):

$$E = \frac{mV_{\mathrm{rel}}^2}{\beta} + mc^2\beta = \frac{mc^2\alpha^2}{\beta}. \quad (52)$$

Use equation (9) to obtain:

$$E = \alpha mc^2\sqrt{1 + \frac{u_{\mathrm{p}}^2}{c^2}} = \alpha\gamma_{\mathrm{local}}mc^2. \quad (53)$$

The energy measured by a local static observer is $E_{local} = E/\alpha = \gamma_{local} mc^2$. At fixed location it diverges as v_{local} approaches c from below for $m > 0$. In flat unloaded space the same action gives $p = \gamma mv$ and $E_{kin} = (\gamma - 1)mc^2$. Thus the limiting-speed energy cost follows from the adopted dynamics.

The conservation check can be shown directly. Metric compatibility gives $d(u_p^2)/d\tau = 2u_{pi}Du_p^i/d\tau$. Contracting equation (51) gives:

$$\frac{d(u_{\mathrm{p}}^2)}{d\tau} = -2(c^2 + u_{\mathrm{p}}^2)\frac{d\ln\alpha}{d\tau}. \quad (54)$$

Consequently:

$$\frac{d\ln(E/(mc^2))}{d\tau} = \frac{d\ln\alpha}{d\tau} + \frac{1}{2(c^2 + u_{\mathrm{p}}^2)}\frac{d(u_{\mathrm{p}}^2)}{d\tau} = 0. \quad (55)$$

The reference energy stays constant while the probe's local kinetic energy and field contribution change along its path. This checks that the adopted motion law conserves energy; the action and spatial response supplied the dynamics.

Expanding equation (53) for $S^2/c^2 \ll 1$ and $v_{local}^2/c^2 \ll 1$ gives:

$$E - mc^2 \approx \frac{1}{2}mv_{\mathrm{local}}^2 - \frac{1}{2}mS^2. \quad (56)$$

For a spherical source this becomes $E_{kin} + U_{probe} \approx \tfrac{1}{2}mv^2 - GMm/r$. Define $\Psi = S^2/2$, the positive energy depth per unit mass. The conventional attractive potential is $\Phi = -\Psi$, so $U_{probe} = -m\Psi$. I retain S^2 for the capacity budget and use Ψ where it simplifies the energy calculation.

If the sources move, gravitational binding energy can change as the system exchanges energy among motion, internal processes, the field and radiation. Conservation applies to the complete energy balance. The source sum $\Sigma Q_a^2/V_a = 24\pi G\Sigma M_a$ has units m^3/s^2 and supplies a separate mass normalization. A collective law must account for both.

4 Clock normalization, propagation and the orbital gradient

4.1 The receiving clock and normalized burden

The same screen readout can be stated per held-clock second. Keeping its spatial normalization unchanged, define

$$Q_\star \, d\tau_{\mathsf{h}} = Q_0 \, dT, \quad Q_\star = \frac{Q_0}{\alpha}. \quad (57)$$

The solved exterior then gives the normalized expression used in the earlier SET algebra:

$$Q_\star(r) = 4\pi r^{3/2} \sqrt{\frac{2GM}{1 - 2GM/(rc^2)}}. \quad (58)$$

The physical source charge has not changed. A slower receiving clock assigns a larger rate to the same specified readout. If a starting rate were defined per source proper time instead of per T, its conversion would also include the source clock factor.

Introduce a speed-valued normalized burden, distinguished from the local capacity burden S:

$$s_\star \equiv \frac{Q_\star}{A_\Sigma} = \frac{S}{\alpha}. \quad (59)$$

The held capacity relation alone gives the following exact identities:

$$\frac{s_\star^2}{c^2} = \frac{1 - A}{A}, \quad A = \frac{1}{1 + s_\star^2/c^2}. \quad (60)$$

$$S^2 = \frac{s_\star^2}{1 + s_\star^2/c^2}. \quad (61)$$

In the solved spherical profile, this becomes

$$s_\star^2 = \frac{2GM/r}{1 - R_s/r} = \frac{2GM}{r - R_s}. \quad (62)$$

This expression is not a new areal radius or a replacement for S in the invariant. Its denominator comes from the receiving clock's normalization. Section 5.1 derives its role in an equivalent orbital energy equation rather than treating it as an additional force.

For a moving receiving clock the same rate bookkeeping uses $Q_{\mathsf{rec}} = Q_0/\beta$. The factors α and β coincide only for a held observer. No literal transport or Doppler interception law is inferred solely by this change of clock units.

This is Q expressed through the proper observer's event rate. For example, at α = 0.8 one proper second spans 1.25 reference seconds, giving Qnormalized = 1.25Q_0. The source QΩ, its strength μQ and the reference-time Q_0 are unchanged by this clock conversion.

Define the dimensionless readout z = Qnormalized/(AΣc), with AΣ = 4πr². Since Q_0/AΣ = S, we have z² = u/α². Substitute α² = 1 − u and rearrange: z²(1 − u) = u, so u(1 + z²) = z². Therefore:

$$\alpha^2 = \frac{1}{1 + z^2}, \quad u = \frac{z^2}{1 + z^2}. \quad (63)$$

Substituting equations (43) and (40) gives equation (58). To recover the occupancy, use u = z²/(1 + z²), or S = cz/√(1 + z²). The static exterior therefore has 0 ≤ u < 1 and 0 ≤ S < c. Qnormalized/AΣ includes the observer's clock normalization; it is not a local material or signal velocity.

4.2 Finite burden differences

Consider two exterior locations $r_1 < r_2$. Before setting $\mathscr{H} = 1$, their separation in the work-current measure is

$$\Delta\chi = \chi_2 - \chi_1 = \int_{r_1}^{r_2} \sqrt{\mathscr{H}(A(r))}\, dr > 0. \quad (64)$$

Subtracting the capacity invariant at the two locations gives an exact acceleration-valued difference after division by $2\Delta\chi$:

$$\frac{S_1^2 - S_2^2}{2\Delta\chi} = \frac{V_{\mathsf{rel},2}^2 - V_{\mathsf{rel},1}^2 + c^2(\beta_2^2 - \beta_1^2)}{2\Delta\chi}. \quad (65)$$

For held readings, $V_{\mathsf{rel},i} = 0$ and $\beta_i = \alpha_i$. Define the outward interval current by

$$\mathscr{J}_{12} \equiv \frac{S_1^2 - S_2^2}{2\Delta\chi} = \frac{c^2(\alpha_2^2 - \alpha_1^2)}{2\Delta\chi}. \quad (66)$$

Its connection to the physical current uses the work relation of Section 3.3. Integrating $\mathscr{J} = -\frac{1}{2} dS^2/d\chi$ shows that $\mathscr{J}_{12}$ is the interval average, not in general the local current at either endpoint:

$$\mathscr{J}_{12} = \frac{1}{\Delta\chi} \int_{\chi_1}^{\chi_2} \mathscr{J}(\chi)\, d\chi. \quad (67)$$

Now express the same burden difference through $s_{\star i} = S_i/\alpha_i$. Applying equation (61) separately at the two endpoints gives

$$S_1^2 - S_2^2 = \frac{c^4(s_{\star 1}^2 - s_{\star 2}^2)}{(c^2 + s_{\star 1}^2)(c^2 + s_{\star 2}^2)}. \quad (68)$$

Because $\alpha_i^2 = c^2/(c^2 + s_{\star i}^2)$, the exact conversion is

$$\mathscr{J}_{12} = \alpha_1^2 \alpha_2^2 \frac{s_{\star 1}^2 - s_{\star 2}^2}{2\Delta\chi}. \quad (69)$$

The product of the two endpoint factors is fixed by the algebra. No midpoint approximation or new coefficient has entered. For positive held capacity the conversion is reversible. The factor $1/2$ and the distance measure have their physical meaning from the retained work coupling; the finite identity itself follows from the capacity relation and the definitions.

4.3 The differential work current

Take the differential limit, or differentiate equation (61) directly:

$$\frac{dS^2}{d\chi} = \frac{c^4}{(c^2 + s_\star^2)^2} \frac{ds_\star^2}{d\chi} = \alpha^4 \frac{ds_\star^2}{d\chi}. \quad (70)$$

The physical work-current can therefore be written in either burden variable:

$$\mathcal{J} = -\frac{1}{2}\frac{dS^2}{d\chi} = -\frac{\alpha^4}{2}\frac{ds_\star^2}{d\chi} = -\frac{\alpha^4}{2\sqrt{\mathscr{H}}}\frac{ds_\star^2}{dr}. \quad (71)$$

This identity holds before response selection. In general $\Delta\chi$ is not $r_2 - r_1$. Once Section 3.4 has selected $\mathscr{H} = 1$, the two radial increments coincide and the solved spherical profile gives

$$\frac{ds_\star^2}{dr} = -\frac{2GM}{(r - R_s)^2}, \quad \alpha^4 = \frac{(r - R_s)^2}{r^2}. \quad (72)$$

The apparent enhancement in the normalized gradient cancels exactly in the physical current:

$$\mathcal{J} = -\frac{1}{2}\frac{(r - R_s)^2}{r^2}\left(-\frac{2GM}{(r - R_s)^2}\right) = \frac{GM}{r^2}. \quad (73)$$

This is an exact exterior identity for $r > R_s$, not a weak-field expansion. For two finite radii the corresponding check is

$$\mathcal{J}_{12} = \frac{2GM/r_1 - 2GM/r_2}{2(r_2 - r_1)} = \frac{GM}{r_1 r_2}. \quad (74)$$

The same value follows from equation (69). It is the average of GM/r^2 over the radial interval and tends to the local current as the endpoints coincide.

These relations connect the invariant-derived burden difference to the current used in the field calculation. They express one current in two normalizations; its conservation across complete vacuum screens remains the physical field law stated in Section 3.3. The held factors cannot simply be replaced by moving-clock factors β_i: the velocity terms in equation (65) must then be retained.

The orbital use of $s_\star$ in Section 5.1 also changes the trajectory parameter and retains the radial kinetic coefficient. Its potential gradient must be interpreted together with those changes. Appendix H records the corresponding exact conversion for squared gradient expressions.

4.4 Light propagation and escape

In the selected exterior, the radial motion coefficient has a compact expression in normalized burden:

$$\zeta^2 = \frac{1}{A} = 1 + \frac{s_\star^2}{c^2}. \quad (75)$$

Its equality to $1/A$ follows from Section 3.4. The identities of Section 4.1 express that selected response through $s_\star$; they do not select it a second time.

Use the retained propagation coupling in which a radial light signal has zero clock progression in the same interval. Setting $\beta = 0$ in the radial capacity equation yields

$$\left|\frac{dr}{dT}\right| = \frac{c\alpha}{\zeta} = cA = \frac{c}{1 + s_\star^2/c^2}. \quad (76)$$

$$\frac{d\ell}{dT} = c\alpha, \quad \frac{d\ell}{d\tau_{\mathsf{h}}} = c. \quad (77)$$

The local clock and the signal propagation participate in the same response. For two nearby radial locations, the reference-clock round-trip time is

$$dT_{\text{round}} = \frac{2\,|dr|}{cA} = \frac{2\,|dr|}{c}\left(1 + \frac{s_\star^2}{c^2}\right). \quad (78)$$

The local radar distance obtained with the emitter's held clock is $c\alpha\, dT_{\text{round}}/2 = |dr|/\alpha$. A finite path requires integration through its varying capacity; the clock at a fixed emitter multiplies the complete reference-time interval by that emitter's lapse.

The distinction between S and $s_\star$ is also fixed by the escape calculation. The probe energy from Section 3.3 can be written

$$\varepsilon \equiv \frac{E}{m_p c^2} = \frac{A}{\beta} = \alpha\gamma, \quad \gamma = \frac{\alpha}{\beta}. \quad (79)$$

Marginal escape to rest at infinity has $\varepsilon = 1$. For the local speed measured by a held observer, this gives

$$\gamma_{\text{esc}} = \frac{1}{\alpha}, \quad v_{\text{esc}}^2 = c^2(1 - \alpha^2) = S^2. \quad (80)$$

Thus S remains the local marginal escape speed in this branch. The normalized quantity is $s_\star = \gamma_{\text{esc}} v_{\text{esc}}$; it is not another local light-speed limit. Appendix D gives the optional free-fall representation after the response has been selected.

The observer's clock participates in the field it is used to measure. In SET, its ticks are physical events, and their progression depends on the available field interaction capacity. Gravitational burden therefore enters both the propagation of light and the clock used to time it.

A local static observer measures radial distance $d\ell r = |dr|/\alpha$ and records time $dt_{\text{local}} = \alpha dT$. Combining these relations with equation (76) gives:

$$\frac{d\ell_r}{dT} = c\alpha, \quad \frac{dt_{\text{local}}}{dT} = \alpha. \quad (81)$$

Light covers less local distance per reference second where α is smaller. The local clock also records fewer ticks per reference second, by the same factor. Consequently, the observer measures:

$$\frac{d\ell_r}{dt_{\text{local}}} = \frac{c\alpha}{\alpha} = c. \quad (82)$$

The common factor cancels in the local measurement. The gravitational effect remains accessible through comparisons between clocks at different burdens and through signal travel times expressed in the reference time T. SET interprets this consistency between propagation and clock progression as their participation in the same limited field interaction capacity. The radial coordinate speed in equation (76) includes both the clock and spatial relations.

For fixed screens at r_1 and r_2, the radial round-trip coordinate time is:

$$\Delta T = \frac{2}{c}\left[r_2 - r_1 + R_s \ln\left(\frac{r_2 - R_s}{r_1 - R_s}\right)\right], \quad R_s = \frac{2GM}{c^2}. \quad (83)$$

The excess over $2(r_2 - r_1)/c$ for solar screens at 1 and 2 AU is approximately 13.656 μs. This is an ideal coordinate-time comparison. A clock at the first screen records $\alpha(r_1)\Delta T$, and its comparison with a chosen flat reference must include that clock factor. The example isolates propagation delay without requiring a bent spatial ray.

S therefore determines the marginal local escape speed in this spherical branch. In the weak field, the energy condition $\frac{1}{2}v_{\text{escape}}^2 - \frac{1}{2}S^2 = 0$ gives the same result. In SET's interpretation the source establishes the burden and energy depth; the escape speed follows from them.

4.5 Dilution and the inward response

Move a probe farther from the spherical source. Doubling r reduces S by $1/\sqrt{2}$ and S^2 by 1/2. The gradient falls faster, by 1/4. To obtain the pull, we must calculate that spatial change.

For a slowly moving probe in a weak static field, $\beta \approx 1$, q_{ij} is approximately Euclidean, and equation (50) becomes:

$$\boldsymbol{a} = \frac{1}{2}\nabla S^2 . \quad (84)$$

Although S^2 is positive, its radial derivative is negative. With $S^2 = 2GM/r$:

$$\nabla S^2 = -\frac{2GM}{r^2}\hat{r}, \quad \boldsymbol{a} = -\frac{GM}{r^2}\hat{r} . \quad (85)$$

The burden is greater on the inner side, so its gradient points inward. Equivalently, α increases outward and $a \approx -c^2\nabla\ln\alpha$ points toward the slower held-clock rate. The energy gives the same direction: $a = -\nabla(U_{probe}/m) = \nabla\Psi$.

I picture this as a difference in event progression across the body. The variational calculation specifies its center-of-mass response without assigning an uncalculated thrust to its internal processes. Tides and internal stresses require an extended-body treatment.

A balance point helps separate burden from pull. Place two equal masses m at distance d on opposite sides of a midpoint. Their ordinary summed potential gives $S^2 = -2\Phi = 4Gm/d$ there, while symmetry makes the gradient zero. The burden remains even though the accelerations balance. This distinction is already present in ordinary weak-field gravity. General Lagrange points also involve rotating-frame terms. The additional collective response proposed below requires its own source-arrangement rule.

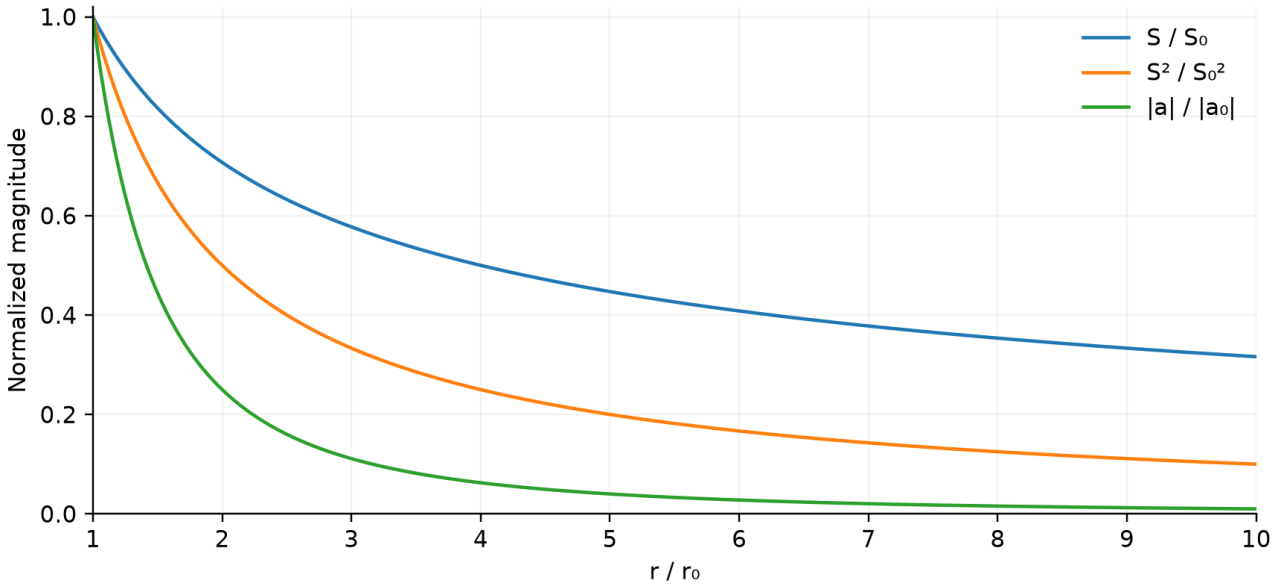


Figure 2. Exterior spherical scalings normalized at r_0. The source equation gives $S \propto r^{-1/2}$ and $S^2 \propto r^{-1}$; the motion law gives $|a| \propto r^{-2}$.

4.6 Orbital speed and the local gradient

Now ask what holds a probe in a circular orbit. The orbit responds to the local gradient of the same field. In a weak, stationary, axisymmetric field, the equatorial radial acceleration is $-v_{orb}^2/R$, giving:

$$v_{orb}^2 = -\frac{R}{2}\frac{\partial S^2}{\partial R} . \quad (86)$$

Define the local equatorial logarithmic slope k(R), with $S^2 > 0$:

$$k(R) = -\frac{\partial \ln S^2}{\partial \ln R}, \quad v_{\mathrm{orb}}^2 = \frac{k(R) S^2}{2}. \quad (87)$$

For the inverse-radius spherical field k = 1. The familiar conversion follows:

$$v_{\mathrm{orb}} = \frac{S}{\sqrt{2}}, \quad \frac{Q_{\mathrm{ref}}^2}{V_{\mathrm{ref}} R} = 24\pi v_{\mathrm{orb}}^2, \quad R = \frac{GM}{v_{\mathrm{orb}}^2}. \quad (88)$$

This derives the speed conversion for the spherical inverse-radius field. More generally, equation (87) shows precisely when it applies: the local logarithmic slope must be k = 1. I use it as the orbital calibration for the collective boundary calculation in Section 7.3. A weak field with another slope must be evaluated through equation (86).

More generally, with S^2 tending to zero at infinity, integrating that weak-field gradient along the equatorial radius gives:

$$S^2(R) = 2 \int_R^{\infty} \frac{v_{\mathrm{orb}}^2(r)}{r} \, dr. \quad (89)$$

Here $\mathrm{vorb}^2(r)$ denotes the circular-speed function supplied by the radial force, wherever such an orbit is defined. The integral requires a convergent exterior continuation. It connects the potential depth at a boundary to the force outside it, rather than determining that depth from one velocity measurement alone. Appendix G illustrates a finite flat-speed region joined to an inverse-radius exterior.

Over a measured interval the same relation can be written without specifying the unobserved exterior:

$$S^2(R_1) - S^2(R_2) = 2 \int_{R_1}^{R_2} \frac{v_{\mathrm{orb}}^2(r)}{r} \, dr. \quad (90)$$

Thus circular-speed observations constrain burden differences under this motion law. They do not alone determine its absolute value. If a boundary value $S^2(R_1)$ is proposed independently, it must leave a nonnegative $S^2(R_2)$ after this subtraction, with an admissible continuation beyond R_2. This is a necessary consistency check, not a sufficient source solution.

5 Perihelion from the spherical field

5.1 Orbital energy in normalized burden

We now derive the normalized-burden form of the motion equations. Temporarily leave $\zeta(r)$ explicit so that the transformation can be checked without presupposing the selected response. The action has no explicit dependence on T or φ. Its constants are

$$\varepsilon = \frac{A}{\beta}, \quad j = \frac{r^2\dot{\varphi}}{\beta} = r^2\frac{d\varphi}{d\tau}. \quad (91)$$

The parameter j is angular momentum per unit probe mass. Introduce an auxiliary parameter along this fixed-energy orbit:

$$d\sigma = \varepsilon\, d\tau = A\, dT, \quad \ell_\sigma = \frac{j}{\varepsilon} = r^2\frac{d\varphi}{d\sigma}. \quad (92)$$

σ is a computational parameter. It is neither the reference clock T, the held clock τ_h nor the probe clock τ. Since ε is constant, it is a constant rescaling of the probe's proper time along this orbit.

Divide the capacity equation by A^2 and use $\beta = A/\varepsilon$:

$$\frac{1}{\varepsilon^2} = \frac{1}{A} - \frac{1}{c^2}\left[\zeta^2\left(\frac{dr}{d\sigma}\right)^2 + r^2\left(\frac{d\varphi}{d\sigma}\right)^2\right]. \quad (93)$$

Now use $s_\star^2 = c^2(1/A - 1)$ and rearrange:

$$\frac{1}{2}\left[\zeta^2\left(\frac{dr}{d\sigma}\right)^2 + r^2\left(\frac{d\varphi}{d\sigma}\right)^2\right] - \frac{s_\star^2}{2} = \mathscr{E}_\sigma, \quad (94)$$

$$\mathscr{E}_\sigma = \frac{c^2}{2}(1 - \varepsilon^{-2}), \quad \Phi_\sigma = -\frac{s_\star^2}{2}. \quad (95)$$

$\mathscr{E}_\sigma$ is an auxiliary conserved energy per mass; physical reference energy remains $E = m_p c^2 \varepsilon$. The original radial coefficient is still present. In the selected spherical field, the effective potential is

$$\Phi_\sigma(r) = -\frac{GM}{r - R_s}. \quad (96)$$

We have obtained this potential variable by transforming the retained action's conserved quantities. To establish full equivalence, the next step verifies its motion equation, not only its energy integral.

5.2 Equivalence of the motion equations

For compact notation on this page, write $q = \zeta^2$, $r_\tau = dr/d\tau$ and $r_{\tau\tau} = d^2r/d\tau^2$; a subscript r denotes a field derivative. Variation of L_p gives

$$\frac{d}{dT}\left(\frac{q\dot{r}}{\beta}\right) = -\frac{c^2 A_r}{2\beta} + \frac{q_r\dot{r}^2}{2\beta} + \frac{r\dot{\varphi}^2}{\beta}. \quad (97)$$

Use $d/dT = \beta\, d/d\tau$ and $\dot{r} = \beta r_\tau$. Divide by β, expand the derivative on the left, and move one half of the q_r term to that side:

$$q r_{\tau\tau} + \frac{q_r}{2} r_\tau^2 - r\varphi_\tau^2 = -\frac{c^2 A_r}{2\beta^2} = -\frac{c^2 \varepsilon^2 A_r}{2A^2}. \quad (98)$$

Since $d\sigma = \varepsilon\, d\tau$ and ε is constant, each quadratic velocity or second derivative contributes the same factor ε^2. After division,

$$q r_{\sigma\sigma} + \frac{q_r}{2} r_\sigma^2 - r\varphi_\sigma^2 = -\frac{c^2 A_r}{2A^2}. \quad (99)$$

Differentiate the normalized-burden identity:

$$\frac{1}{2}\frac{ds_\star^2}{dr} = \frac{c^2}{2}\frac{d}{dr}\left(\frac{1}{A} - 1\right) = -\frac{c^2 A_r}{2A^2}. \quad (100)$$

Equation (71) identifies the relation to the physical work-current, before or after selecting $\mathscr{H}$:

$$\frac{1}{2}\frac{ds_\star^2}{dr} = -\frac{\sqrt{\mathscr{H}}}{\alpha^4}\mathscr{J}. \quad (101)$$

The right side of the auxiliary motion equation is therefore not the physical work-current itself. The conversion to σ and the retained radial kinetic coefficient q belong to the same transformed equation. Multiplying its right side by an additional α^4 would change the orbit rather than complete the normalization.

Equations (99) and (100), together with $r^2\varphi_\sigma = \ell_\sigma$, are the equations of the auxiliary Lagrangian

$$\frac{L_\sigma}{m_p} = \frac{1}{2}(q r_\sigma^2 + r^2\varphi_\sigma^2) + \frac{s_\star^2}{2}. \quad (102)$$

This establishes equivalence of both energy and motion for each orbit. We transformed the original Euler–Lagrange equation before identifying L_σ; a trajectory-dependent substitution of time directly into an action need not preserve its variational problem.

Normalized burden therefore has an exact role as the potential variable for these parameterized trajectories. Its radial derivative and the radial motion coefficient must be used together. Adding a second enhancement to the force because $Q_\star > Q_0$ would count the same clock normalization twice.

5.3 The exact orbit equation

Use the selected response $q = 1/A$ in equation (94). Put $w = 1/r$ and let a prime on w denote $d/d\varphi$. Since $r_\sigma = -\ell_\sigma w'$, the energy equation becomes

$$\frac{\ell_\sigma^2}{A}(w')^2 + \ell_\sigma^2 w^2 = s_\star^2 + c^2(1 - \varepsilon^{-2}). \quad (103)$$

Multiply by A and use $A s_\star^2 = c^2(1 - A)$:

$$\ell_\sigma^2 (w')^2 + A\ell_\sigma^2 w^2 = c^2(1 - A\varepsilon^{-2}). \quad (104)$$

Finally multiply by ε^2, recalling $j = \varepsilon \ell_\sigma$:

$$j^2(w')^2 = c^2\varepsilon^2 - A(c^2 + j^2w^2). \quad (105)$$

This is also obtained directly from the original capacity action using $dT/d\tau = \varepsilon/A$ and $dr/d\tau = -jw'$. The normalized-burden and original-clock routes therefore agree before any weak-field expansion.

Substitute the solved $A = 1 - 2GMw/c^2$ and expand the product:

$$j^2(w')^2 = c^2(\varepsilon^2 - 1) + 2GMw - j^2w^2 + \frac{2GMj^2w^3}{c^2}. \quad (106)$$

Differentiate with respect to φ:

$$2j^2w'w'' = 2GMw' - 2j^2ww' + \frac{6GMj^2w^2w'}{c^2}. \quad (107)$$

Where $w' \neq 0$, divide by $2j^2w'$. The smooth second-order equation extends to isolated turning points by continuity:

$$w'' + w = \frac{GM}{j^2} + \frac{3GMw^2}{c^2}. \quad (108)$$

The coefficient 3 comes from differentiating the cubic term generated by the solved response and the angular-motion term. It has not been adjusted to a measured advance. Circular motion follows directly from the radial Euler–Lagrange equation and does not require division by w'.

5.4 The leading perihelion advance

Define $\ell_g = GM/c^2$ and $p_0 = j^2/(GM)$. The unperturbed solution is

$$w_0 = \frac{1 + e\cos\varphi}{p_0}, \quad w_0'' + w_0 = \frac{1}{p_0}. \quad (109)$$

Write $w = w_0 + \delta w$ and retain first order in ℓ_g/p_0. In the correction term, replace w by w_0:

$$\delta w'' + \delta w = 3\ell_g w_0^2. \quad (110)$$

$$3\ell_g w_0^2 = \frac{3\ell_g}{p_0^2}\left(1 + \frac{e^2}{2} + 2e\cos\varphi + \frac{e^2}{2}\cos 2\varphi\right). \quad (111)$$

The constant and double-frequency terms give bounded changes. The term proportional to $\cos\varphi$ is resonant with the unperturbed radial oscillation. Use

$$\left(\frac{d^2}{d\varphi^2} + 1\right)(\varphi\sin\varphi) = 2\cos\varphi, \quad (112)$$

to obtain a particular correction:

$$\delta w = \frac{3\ell_g}{p_0^2}\left(1 + \frac{e^2}{2} + e\varphi\sin\varphi - \frac{e^2}{6}\cos 2\varphi\right). \quad (113)$$

The homogeneous terms shift the fitted eccentricity and initial phase. To read the accumulated advance, compare the resonant term with

$$\frac{e}{p_0}\cos[(1-\delta)\varphi] \simeq \frac{e}{p_0}\cos\varphi + \frac{e\delta}{p_0}\varphi\sin\varphi\,. \quad (114)$$

$$\delta = \frac{3\ell_g}{p_0},\quad \Delta\varphi = \frac{2\pi}{1-\delta} - 2\pi \simeq \frac{6\pi\ell_g}{p_0}\,. \quad (115)$$

To this order $p_0 = a(1-e^2)$, giving the advance per radial period:

$$\Delta\varphi = \frac{6\pi GM}{a(1-e^2)c^2} + O\left(\frac{G^2M^2}{a^2(1-e^2)^2c^4}\right). \quad (116)$$

The final formula assumes a weak-field bound orbit. Equation (108) itself is exact within the selected stationary spherical test-body model.

5.5 Exact orbit verification

For an exact check, specify the turning radii by $r_p = p/(1+e)$ and $r_a = p/(1-e)$, with $P = p/\ell_g$. Setting equation (105) to zero at each turning point and subtracting the two equations gives

$$\frac{j^2}{GM\ell_g} = \frac{P^2}{P-3-e^2},\quad \varepsilon^2 = \frac{(P-2)^2-4e^2}{P(P-3-e^2)}\,. \quad (117)$$

Here p is defined by the exact turning radii. It differs from $p_0 = j^2/(GM)$ beyond leading order. Substitute $w = (1+e\cos\xi)/p$ into equation (106):

$$(w')^2 = \frac{e^2\sin^2\xi}{p^2}\left[1 - \frac{2(3+e\cos\xi)}{P}\right]. \quad (118)$$

For $P > 6+2e$, cancellation against $(dw/d\xi)^2$ gives the total-angle integral and advance:

$$\Delta\varphi = \int_0^{2\pi}\left[\left(1-\frac{2(3+e\cos\xi)}{P}\right)^{-1/2} - 1\right]d\xi\,. \quad (119)$$

Independent numerical quadrature and direct integration of equation (108) agree. The values below use $e = 0.2$ and express p in units of GM/c^2. They are checks of the equations, not additional astronomical observations.

Table 2. Exact orbit checks at eccentricity 0.2

P	Advance from quadrature rad	Advance from orbit integration rad
100	0.1974475160	0.1974475160
1000	0.0189349966	0.0189349966
10000	0.0018858061	0.0018858061

The same calculation with the alternative radial response is carried through in Appendix B. The normalized energy and motion transformation was also checked with exact rational arithmetic before the numerical integrations.

The inequality specifies the stable bound-orbit domain of this expression. The e = 0 case is the small radial-oscillation limit of a circular orbit. Equation (116) is the first-order expansion of equation (119), using the appropriate weak-field orbital elements.

5.6 The Mercury calculation

Use the solar gravitational parameter and constants from NASA JPL [12], together with JPL's approximate J2000 Mercury elements [9]:

$$GM_{\odot} = 1.3271244004127942 \times 10^{20}\ \mathrm{m^3/s^2}, \quad (120)$$

$$c = 299792458\ \mathrm{m/s}, \quad 1\ \mathrm{au} = 149597870700\ \mathrm{m}, \quad (121)$$

$$a = 0.38709927\ \mathrm{au}, \quad e = 0.20563593\,. \quad (122)$$

These give the orbital scale and expansion parameter

$$p = a(1 - e^2) = 5.546046913 \times 10^{10}\ \mathrm{m}, \quad (123)$$

$$\frac{GM_{\odot}}{pc^2} = 2.662482056 \times 10^{-8}\,. \quad (124)$$

Equation (116) then gives

$$\Delta\varphi_{\mathrm{orbit}} = 5.018660440 \times 10^{-7}\ \mathrm{rad} = 0.1035173023\ \mathrm{arcsec}\,. \quad (125)$$

With 36525 days per Julian century [12] and the listed sidereal period 87.969 days [13],

$$\Delta\varphi_{\mathrm{century}} = 0.1035173023\frac{36525}{87.969}\ \mathrm{arcsec} = 42.98070\ \mathrm{arcsec}\,. \quad (126)$$

This is the standard approximately 43 arcseconds per century spherical relativistic contribution [14]. The source and response coefficients were fixed before this numerical substitution. The approximate elements are benchmark inputs, not a new fit to Mercury's observations; their precision does not support interpreting all displayed digits as empirical accuracy.

The calculation neglects the other planets, source rotation and multipoles, and test-body backreaction. Recovering the spherical contribution does not constitute a calculation of Mercury's complete observed precession. Appendix C uses the same inputs to evaluate the normalized readout during the finite time spent near perihelion.

The earlier calculation retained with Draft 48 uses the slightly different period 87.96946593 days. It reports 42.98048057 arcseconds per Julian century from exact quadrature. The main benchmark above uses the listed 87.969-day convention. Both give the same approximately 42.98 arcsecond spherical contribution; the period convention and the exact-versus-leading comparison are retained with the numerical checks in Appendix I.

6 Strong fields and rotating black hole exteriors

The capacity calculation can also be carried into a stationary strong field. I summarize the source-cap and rotating construction here because the same framework should be followed into demanding calculations, not restricted to weak fields. The full derivation and photon calculation are given in the companion black-hole paper [29].

6.1 The source cap and rotation

$$r_s = \frac{2GM}{c^2}, \quad Q_H(0) = 4\pi r_s^2 c = \frac{16\pi G^2 M^2}{c^3}. \quad (127)$$

The horizon screen reading is finite. Its held-clock normalization QH/α diverges as the stationary held-clock limit is approached; those are different readings. Two masses can both have S = c at their respective caps while establishing different exterior profiles, $S(r_s + d) = c\sqrt{[r_s/(r_s + d)]}$.

$$S^2 + \mathscr{C}_\varphi(\Omega - \omega)^2 + c^2\beta^2 = c^2. \quad (128)$$

At fixed radial and polar position, Ω is the probe angular rate, ω is the local frame rate and $\mathscr{C}\varphi$ is the squared circumferential lever. Rotation therefore enters through the motion term. To connect that algebra to a source state, let $AH = 4\pi RA^2$ and $mA = RAc^2/(2G)$, and retain the source-energy relation below. It is the uncharged mass-area relation in conventional black-hole variables [23].

$$M^2c^4 = m_A^2c^4 + \frac{c^2J^2}{R_A^2}, \quad \mu = \frac{GM}{c^2}, \quad a_J = \frac{J}{Mc}. \quad (129)$$

$$r_H^2 + a_J^2 = R_A^2 = 2\mu r_H, \quad r_H = \mu + \sqrt{\mu^2 - a_J^2}. \quad (130)$$

$$Q_H(M, J) = \frac{8\pi G^2 M^2}{c^3}\left(1 + \sqrt{1 - \frac{c^2J^2}{G^2M^4}}\right). \quad (131)$$

For $|cJ/(GM^2)| \le 1$, this returns the nonrotating value at J = 0 and approaches half that value in the extremal limit at fixed total mass. These are different source states with different horizon areas.

6.2 From the comoving source screen to motion

The retained exterior uses confocal oblate reference screens, their local comoving area, and the reciprocal normal propagation response. With $F = r^2 + aJ^2$, their reference volume is $V_0 = 4\pi rF/3$ and their comoving area is $Acom = 4\pi F$. The source law then gives

$$Q_{\text{ref}}^2 = 24\pi GMV_0, \quad U^2 = \left(\frac{Q_{\text{ref}}}{A_{\text{com}}}\right)^2 = \frac{2GMr}{F}. \quad (132)$$

Here U is the comoving screen burden. The external normal-frame burden S is obtained when the completed interval is reorganized into its angular square; the two readings coincide at the cap and in the nonrotating limit. Define $\Sigma = r^2 + aJ^2\cos^2\theta$ and $\Delta = F - 2\mu r$. The source and normal-response correspondences give

$$c^2 d\tau^2 = \frac{\Delta}{\Sigma}(c\,dT - a_J \sin^2\theta\, d\varphi)^2 - \frac{\Sigma}{\Delta} dr^2 - \Sigma\, d\theta^2$$

$$-\frac{\sin^2\theta}{\Sigma}(F\,d\varphi - a_J c\,dT)^2 . \quad (133)$$

The two displayed lines are one interval. Expanding it gives the Kerr exterior [24]. The source-energy identification, comoving screen correspondence and normal response specify this realization; the known rotating target informed its construction. Applying the same calibrated probe action gives the equatorial circular rates

$$\Omega_{\pm} = \frac{\pm c\sqrt{\mu / r^3}}{1 \pm a_J \sqrt{\mu / r^3}} . \quad (134)$$

The nonrotating limit returns the spherical orbit. The full calculation also derives radial stability and photon capture from this interval. In the far exterior, ω approaches 2GJ/(c²r³); the corresponding tangential frame speed has one less inverse power of radius because its lever grows with r.

6.3 An observed strong field scale

The companion photon calculation scans 54 spins from a* = cJ/(GM²) = 0 to 0.9999 and 19 viewing inclinations from 0 to 90 degrees. Its sampled shadow diameter is 9.6571–10.3923 times GM/c². Using the stellar-orbit angular scales quoted by the EHT analysis gives 49.49–53.26 microarcseconds for VLTI and 47.51–51.13 for Keck. These are parameter ranges, with source-scale errors treated separately [25].

The reference eht-imaging shadow inference is 48.9 microarcseconds, with a 68 percent interval of 43.8–54.1. It contains both calculated ranges. This comparison uses the calibrated shadow inference, not the distinct 51.8 ± 2.3 microarcsecond emission-ring measurement. The stellar-orbit scales assume GR, and the shadow inference includes emission calibration. The result checks the strong-field scale of the specified Kerr-equivalent SET construction [25, 27].

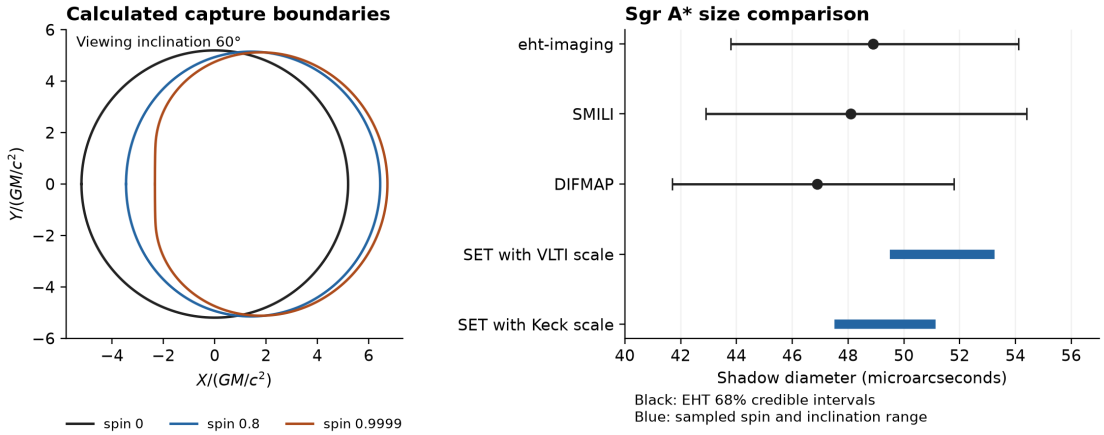


Figure 3. Left: null capture contours calculated from the completed SET exterior at a common inclination. Right: EHT shadow inferences using the GRMHD calibration [25] and our sampled parameter ranges at the central stellar-orbit scales. Blue segments are parameter ranges; black bars are 68 percent credible intervals. The source-scale errors are listed in the companion black-hole paper. No emission brightness model is calculated here.

For a concrete source reading, the quoted VLTI mass 4.297 × 10^6 M⊙ gives QH(0) = 6.067 × 10^{29} m³/s. At a* = 0.8 the rotating expression gives 4.853 × 10^{29} m³/s. These follow from the source mass and chosen spin; the observational comparison is the shadow size calculated through the propagation response [25, 28].

7 Effective area and collective source organization

7.1 Boundary loading and its effective area

I am asking whether the source–boundary relation used for an individual body also applies to a collective system of disjoint masses. At Earth's approximately spherical surface, the source normalization is read through the spherical area. For a galaxy, SET retains the same normalization law and proposes that the organization of its baryonic sources enters through an effective boundary area. The physical question is whether the resulting burden, together with SET's motion law, can account for the observed orbital speeds using the baryonic inventory alone.

There is a geometrical distinction between a point among sources and a boundary enclosing them. For a convex enclosing surface, such as a sphere or an oblate spheroid, every source-to-boundary direction has a nonnegative outward-normal component. In the stationary model developed in Appendix F, the contributions from the enclosed sources to the defined source flux therefore add with the same outward sign at that surface, although their strengths and directions can differ. Between sources, directional contributions can oppose while burden remains present. This is the distinction I have in mind when considering a collective system through an enclosing boundary. I now define an effective area to describe its boundary loading.

For a nonnegative scalar loading s on a specified boundary Σ, define:

$$Q_\Sigma = \int_\Sigma s\, dA, \quad I_\Sigma = \int_\Sigma s^2\, dA\,. \quad (135)$$

$$A_{\mathrm{eff}} = \frac{Q_\Sigma^2}{I_\Sigma}, \quad S_{\mathrm{eff}} = \frac{I_\Sigma}{Q_\Sigma} = \frac{Q_\Sigma}{A_{\mathrm{eff}}}\,. \quad (136)$$

The first moment QΣ has units m^3/s; the second moment IΣ has units m^4/s^2. If s^2/c^2 is the local occupancy under the SET identification, IΣ/(Ageom c^2) is the area-averaged occupancy. Cauchy–Schwarz gives $Q\Sigma^2 \leq$ Ageom IΣ and therefore:

$$A_{\mathrm{eff}} \leq A_{\mathrm{geom}}, \quad \frac{A_{\mathrm{geom}}}{A_{\mathrm{eff}}} = 1 + \frac{\mathrm{Var}(s)}{\langle s \rangle^2}\,. \quad (137)$$

Take QΣ > 0. Uniform loading gives Aeff = Ageom. To see what concentration changes, use four equal unit-area patches. The loadings (1,1,1,1), (2,2,0,0) and (4,0,0,0) all carry QΣ = 4. Their second moments are 4, 8 and 16, so Aeff is 4, 2 and 1. The same total loading occupies a smaller effective area as it becomes more concentrated. Conversely, multiplying all loadings by the same factor leaves Aeff unchanged. The area measures concentration; the normalization fixes the amplitude.

Two surface integrals now appear. $Q\Sigma = \int\Sigma$ s dA has units m^3/s. Appendix F.1's gradient flux $\oint\Sigma\, \mathcal{J}\cdot dA = 4\pi GM$ has units m^3/s^2. Their conservation properties differ. The gradient-flux law leaves the boundary loading normalization undetermined: Appendix F shows that rescaling Ψ and inversely rescaling C preserves $\mathcal{J}$ while changing the force. Section 7.3 supplies the QΣ normalization used for the collective calculation.

If source amplitudes add at one location, their square contains overlap terms. The boundary identity itself leaves that physical superposition rule open. Also, Seff averages the boundary loading with weight s: it equals the local field S on a uniformly loaded boundary, but generally summarizes a distribution. Applying it to probe motion requires a relation to the local S and its gradient.

An effective boundary must describe the physical response, rather than create a response merely because we chose the boundary. Rearranging the sources can change the field distribution; merely redrawing an enclosing surface cannot change the field at a fixed location. I test a specific prescription: retain the spherical reference Qref for the adopted mass and radius, and estimate Aeff with an oblate surface constrained by

imaging. This connects the proposed collective response to observed source geometry. The collective model must determine the relevant boundary or give results independent of an arbitrary boundary choice.

7.2 Spherical reference and image geometry

For a spherical evaluation screen at R, introduce its reference volume $V_{ref} = (4/3)\pi R^3$ and write $Q_{ref} = Q_0(R)$. Equation (43) gives $Q_{ref}^2/V_{ref} = 24\pi GM$. Expanding both quantities makes the normalization explicit:

$$\frac{[4\pi\sqrt{2GMR^3}]^2}{(4/3)\pi R^3} = 24\pi GM\,. \quad (138)$$

Divide both sides by R:

$$\frac{Q_{\mathrm{ref}}^2}{V_{\mathrm{ref}}R} = \frac{24\pi GM}{R} = 12\pi S^2(R)\,. \quad (139)$$

The geometric step is $12\pi V_{ref} R = 16\pi^2 R^4 = (4\pi R^2)^2$. Therefore:

$$S^2(R) = \frac{Q_{\mathrm{ref}}^2}{12\pi V_{\mathrm{ref}}R} = \left[\frac{Q_{\mathrm{ref}}}{4\pi R^2}\right]^2. \quad (140)$$

For positive Q_{ref} and S, taking the positive square root of equation (140) gives $4\pi R^2 S = Q_{ref}$. Solving for R, or using $S^2 = 2GM/R$, gives the equivalent forms:

$$R = \frac{Q_{\mathrm{ref}}^2}{12\pi V_{\mathrm{ref}}S^2} = \sqrt{\frac{Q_{\mathrm{ref}}}{4\pi S}} = \frac{2GM}{S^2}\,. \quad (141)$$

The same spherical relation can therefore be written in terms of radius, amplitude or source normalization. In the first form, V_{ref} still depends on R; substituting $V_{ref} = (4/3)\pi R^3$ gives the square-root form. Q_{ref}^2/V_{ref} has units m^3/s^2, and division by R gives m^2/s^2, the units of S^2 and energy per unit mass. Since Q_{ref} was calculated using R, rearranging it does not independently determine a body's size or internal density. For galaxies I retain this spherical reference volume while changing the boundary area.

I denote this particular solution by $S_{sph}(R) = \sqrt{(2GM/R)}$. S itself is the burden determined by the source arrangement and response. In Section 7.3 I retain the reference Q while allowing the effective area, and therefore the boundary amplitude, to change.

For an oblate spheroid with equatorial radius R, polar radius ηR, and eccentricity $e_\eta = \sqrt{(1 - \eta^2)}$, the geometric surface is:

$$A_{\mathrm{geom}} = f\pi R^2,\;\; f = 2\left[1 + \frac{\eta^2 \operatorname{atanh}(e_\eta)}{e_\eta}\right]. \quad (142)$$

The limits are $f \to 2$ for a thin oblate surface and $f = 4$ for a sphere. A projected minor-to-major ratio p_{image} at inclination i gives, under the axisymmetric oblate assumption:

$$\eta^2 = \frac{p_{\mathrm{image}}^2 - \cos^2 i}{\sin^2 i}\,. \quad (143)$$

This reconstructs an intrinsic oblate shape from a projected image and an adopted inclination. At $i = 0$ the image cannot determine η; near that orientation the inference becomes sensitive to measurement error. Bars, warps, triaxiality and varying inclination can also affect it. I flag a negative central η^2 as unresolved

deprojection rather than clipping it to a thin disk. The range $2 \le f \le 4$ applies to these geometric oblate surfaces; the participation area in equation (136) can also reflect nonuniform loading.

7.3 The weak field boundary calculation

For the collective boundary calculation, take the adopted baryonic mass Mbar and comparison radius R. I retain Vref = (4/3)πR³ as the source reference volume and identify the boundary loading with Qref = Aeff Seff. The source normalization and effective area then determine the boundary amplitude together. The reference quantity and image-based reading area are:

$$Q_{\mathrm{ref}} = 4\pi\sqrt{2GM_{\mathrm{bar}}R^3}, \quad A_{\mathrm{image}} = f_{\mathrm{image}}\pi R^2 . \quad (144)$$

With QΣ = Qref, combine Qref²/Vref = 24πGMbar with Qref = Aeff Seff:

$$\frac{A_{\mathrm{eff}}^2 S_{\mathrm{eff}}^2}{V_{\mathrm{ref}}} = 24\pi GM_{\mathrm{bar}} . \quad (145)$$

Solving directly for the positive boundary amplitude gives:

Seff = √(24πGMbarVref)/Aeff.

For Aeff = fπR², the effective boundary amplitude is therefore:

$$S_{\mathrm{eff}}^2 = \frac{32GM_{\mathrm{bar}}}{f^2 R} . \quad (146)$$

The spherical case f = 4 returns Ssph = √(2GMbar/R). The thin-area limit f = 2 gives Seff = √(8GMbar/R). Both carry the same reference Q:

$$\begin{aligned} Q_{\mathrm{ref}} &= \sqrt{32\pi^2 GM_{\mathrm{bar}}R^3} = 4\pi\sqrt{2GM_{\mathrm{bar}}R^3} \\ &= f\pi\sqrt{\tfrac{32}{f^2}GM_{\mathrm{bar}}R^3} = (f\pi R^2)\sqrt{\tfrac{32GM_{\mathrm{bar}}}{f^2 R}} = A_{\mathrm{eff}}S_{\mathrm{eff}} . \end{aligned} \quad (147)$$

$$4\pi R^2\sqrt{\frac{2GM_{\mathrm{bar}}}{R}} = 2\pi R^2\sqrt{\frac{8GM_{\mathrm{bar}}}{R}} = Q_{\mathrm{ref}} . \quad (148)$$

These are equivalent expressions for the same reference Q at fixed baryonic mass, reference radius and reference volume. In this collective prescription, the effective area and boundary burden change reciprocally through Qref = Aeff Seff. The spherical expression √(2GM/R) is the f = 4 case; it is not held fixed when the effective area changes. Where Seff represents the local S, the energy relation in Section 4.4 gives the corresponding marginal escape speed. This is the physical interpretation proposed by SET: source organization changes the boundary burden, and that burden determines the escape-energy scale. Moving the observation radius through one field is a different operation, for which $Q_0(r)$ follows the field solution. For the observational calculation I approximate Aeff by Aimage and call the resulting amplitude Sboundary:

$$S_{\mathrm{boundary}} = \frac{Q_{\mathrm{ref}}}{A_{\mathrm{image}}}, \quad v_{\mathrm{SET}} = \frac{S_{\mathrm{boundary}}}{\sqrt{2}} . \quad (149)$$

Combining the expressions makes the prediction explicit:

$$v_{\mathrm{SET}} = \frac{4}{f_{\mathrm{image}}}\sqrt{\frac{GM_{\mathrm{bar}}}{R}} . \quad (150)$$

To connect the boundary reading to orbital motion, distinguish the radius Rb of the reference boundary from the equatorial orbital radius R. Write Sb = Sboundary(Rb). For a conditional fixed-mass, fixed-shape inverse-radius exterior, the continuation is

$$S^2(R) = S_b^2 \frac{R_b}{R}, \quad R \geq R_b . \quad (151)$$

If d is distance outward from that boundary, R = Rb + d and

$$S^2(R_b + d) = S_b^2 \frac{R_b}{R_b + d} . \quad (152)$$

$$\frac{dS^2}{dd} = -\frac{S_b^2 R_b}{(R_b + d)^2} . \quad (153)$$

Measuring distance from the boundary does not remove the source's radius from the denominator. The motion law uses the orbital radius R, giving

$$v_{\text{orb}}^2(R) = \frac{S_b^2 R_b}{2R} . \quad (154)$$

At R = Rb this is vorb = Sb/√2. Substituting Sb² = 32GMbar/(f²Rb) gives vorb²(R) = 16GMbar/(f²R): the reference boundary radius cancels. Moving a reference screen through this same fixed-shape exterior therefore leaves the field at a fixed observation point unchanged. This continuation is conditional on the stated exterior profile; selecting an image radius does not impose it on the physical field.

The reference volume used to define Qref is still (4/3)πRb³. The actual oblate enclosed volume is η times that value. Replacing the reference volume by the actual oblate volume in the source formula would change Q by √η and define a different prescription. Equations (144)–(145) state which normalization is being tested.

For an arbitrary specified source distribution, I can check the approximation against the solved local field. Define vB(R) = (4/f)√(GMbar/R), SB(R) = √2vB(R), and b_s(R) = Slocal(R,0)/SB(R). With k defined in equation (87), the motion law gives

$$\frac{v_{\text{field}}}{v_B} = b_s \sqrt{k} . \quad (155)$$

The boundary estimate's fractional difference from that field prediction is therefore $|1/(b_s\sqrt{k}) - 1|$, wherever the radial force is inward. These are diagnostics of the field solution, not adjustable response factors. The stated fixed-mass, fixed-shape inverse-radius continuation has $b_s = k = 1$. For a more general source, the same calculation tests whether the boundary estimate is accurate enough for its stated use. A prospective application should specify its approximation tolerance before inspecting observed-speed agreement, then assess it over a stated radial neighborhood and allowed source reconstructions. Identifying SB with the local escape burden additionally requires b_s to be close to one; amplitude and slope deviations can compensate in the speed.

The galaxy protocol tests the identification of the image area with Aeff and the use of Sboundary in this orbital conversion. The second-moment identities quantify a known loading; they do not establish that identification from an image. Equation (150) states the particular prescription used for the comparisons.

7.4 Inverse readings of the same relation

To express the same boundary normalization in orbital variables, use f = fimage and Qref = fπR²Sboundary. Dividing the squared relation by Vref R gives:

$$\frac{Q_{\mathrm{ref}}^2}{V_{\mathrm{ref}}R} = \frac{3\pi f^2}{4} S_{\mathrm{boundary}}^2 . \quad (156)$$

With the orbital calibration Sboundary² = 2vSET², the same expression becomes:

$$\frac{Q_{\mathrm{ref}}^2}{V_{\mathrm{ref}}R} = \frac{3\pi f^2}{2} v_{\mathrm{SET}}^2 = \frac{24\pi G M_{\mathrm{bar}}}{R} . \quad (157)$$

Canceling common factors gives vSET² = 16GMbar/(f²R), consistent with equation (150). Its inverse forms are:

$$R = \frac{16 G M_{\mathrm{bar}}}{f^2 v_{\mathrm{SET}}^2}, \quad M_{\mathrm{required}} = \frac{f^2 v_{\mathrm{obs}}^2 R}{16 G} . \quad (158)$$

For f = 4, the inverse relation is R = GMbar/vSET². At the same mass and radius, f = 2 doubles the predicted speed relative to the spherical reference. Here R is the orbital comparison radius. Its algebraic inversion does not independently locate the source's physical edge: the reference boundary radius cancels from the fixed-shape exterior relation. A measured speed and orbital radius constrain Mbar/f² under this prescription. An independently constrained mass permits inference of f, and an independently constrained f permits inference of the required mass. These are conditional inferences to be checked against the source observations. A Newtonian disk calculation uses the actual spatial mass distribution, as shown in Section 8.4.

7.5 Observational inputs and comparison radii

At vSET = 200 km/s, the prescription gives Sboundary = 282.84 km/s and Sboundary²/c² ≈ 8.90×10⁻⁷. These are weak-field values; relativistic corrections of that scale cannot account for several-percent speed discrepancies. The orbital slope condition of Section 4.6 still applies.

Across a range of radii, constant Mbar and f would give vSET ∝ R$^{-1/2}$. A flat circular-speed profile instead requires S² = S₀² − 2vflat² ln(R/R₀) over that range, where S₀² = S²(R₀). Appendix G joins such a finite region to an inverse-radius exterior. The galaxy table tests the prescription at specified radii; deriving the field across the measured curve is a further calculation.

For the nineteen SPARC catalogue cases in the selected set, use [5]:

$$M_{\mathrm{bar}} = 0.5 L_{\mathrm{disk}} + 0.7 L_{\mathrm{bulge}} + 1.33 M_{\mathrm{HI}} . \quad (159)$$

with luminosities in solar 3.6-micron units, stellar mass-to-light ratios in their corresponding solar units, and the result in solar masses. The factor 1.33 accounts for the adopted helium convention. This inventory omits molecular gas and is not a complete enclosed census at every interior radius. The supplementary NGC 4565 calculation uses the same mass recipe with the separate published photometric decomposition and HI flux [7, 8].

The comparison radius is the outermost actual rotation-curve point bracketed by qualifying image isophotes. The rule does not search radii for the smallest velocity discrepancy. Recalculate Qref at that radius. The image pipeline retains positive intensity, finite formal ellipse errors no larger than 0.1 in axis ratio, and surface-brightness signal-to-noise at least 3. It interpolates between bracketing valid isophotes and uses the larger bracketing formal shape error. Fine and coarse samplings are recorded separately [6].

This rule locates a usable comparison point. Whether that point admits the collective exterior approximation is a separate question about the source and its field. A galaxy need not have a sharply terminated density profile for the approximation to be useful; its accuracy is what matters.

An inferred area coefficient is useful as a diagnostic:

$$f_{\text{inferred}} = \frac{4\sqrt{GM_{\text{bar}}/R}}{v_{\text{obs}}}. \quad (160)$$

This inferred coefficient can be compared with the independently reconstructed image coefficient. A value outside $2 \le f \le 4$ cannot describe the geometric oblate surface of equation (142) at that radius. The more general participation area in equation (136) can be smaller under nonuniform loading, but invoking that possibility requires a corresponding field-loading calculation; it is not an image measurement of a more flattened spheroid.

Its agreement with the image coefficient is exactly related to speed agreement:

$$\frac{f_{\text{inferred}}}{f_{\text{image}}} = \frac{v_{\text{SET}}}{v_{\text{obs}}}. \quad (161)$$

8 Galaxy calculations and observed velocities

8.1 The twenty examples

The question in these examples is whether the area estimated from a galaxy's image corresponds to the boundary amplitude required to account for its observed orbital speed. I present twenty worked applications chosen to illustrate close agreement under the stated prescription. In each calculation, the adopted baryonic mass, comparison radius and image-derived area determine the SET speed; no separate halo mass or concentration is adjusted. The interest is the connection between an observed property of the source arrangement and the gravitational response proposed by SET. The research supplement retains the broader calculation record and controls.

Galaxy	R kpc	Mbar 10^{10} M⊙	f image	f inferred	v observed km/s	v SET km/s	Difference
NGC4051	12.19 E	5.122	3.422	3.515	153.0 ± 9.8	157.1	+2.71%
NGC5371	23.12 I	18.507	3.049	3.092	240.0 ± 3.0	243.4	+1.42%
UGC07690	2.36 I	0.095	2.664	2.827	58.8 ± 4.3	62.4	+6.13%
NGC3949	7.07 E	2.352	2.589	2.831	169.0 ± 25.5	184.8	+9.34%
NGC4013	13.47 I	4.657	2.557	2.637	185.0 ± 6.0	190.7	+3.09%
UGC06818	5.24 I	0.223	2.551	2.403	71.2 ± 5.2	67.1	-5.80%
NGC2683	11.54 I	4.315	2.371	2.536	200.0 ± 5.0	213.9	+6.97%
NGC3198	16.07 I	3.360	2.321	2.479	153.0 ± 2.0	163.4	+6.81%
NGC3992	27.52 I	13.554	2.255	2.205	264.0 ± 5.3	258.2	-2.21%
UGC08837	4.20 E	0.068	2.217	2.193	48.0 ± 3.0	47.5	-1.10%
NGC5585	6.16 I	0.371	2.201	2.313	88.0 ± 3.0	92.5	+5.13%
NGC4010	10.47 E	1.236	2.183	2.337	122.0 ± 3.9	130.6	+7.05%
NGC4565	19.75 E	6.540	2.182	2.104	226.9 ± 22.0	218.8	-3.58%
NGC4183	11.35 I	1.008	2.181	2.248	110.0 ± 5.6	113.3	+3.04%
NGC5907	27.68 I	11.568	2.178	2.342	229.0 ± 2.0	246.2	+7.51%
NGC0100	8.90 I	0.426	2.164	2.040	89.0 ± 7.8	83.9	-5.76%
NGC4100	13.94 I	3.382	2.163	2.245	182.0 ± 4.5	188.9	+3.81%
NGC5033	24.01 I	7.533	2.126	2.093	222.0 ± 2.0	218.5	-1.57%
UGC07089	9.16 E	0.341	2.117	2.023	79.1 ± 7.3	75.6	-4.46%
NGC1090	23.89 I	4.770	2.112	2.247	165.0 ± 2.2	175.5	+6.35%

Table 3. Twenty single-radius comparisons. In the radius column E denotes the endpoint of the adopted rotation dataset and I an interior comparison point. These labels describe the extent of the adopted velocity dataset, not the physical source boundary. Neither an endpoint nor an earlier measured orbit, by itself, establishes the local field's exterior character. Mbar is the adopted global stellar-plus-HI/helium inventory, in 10^{10} M⊙. The observed speeds predominantly trace gas rotation. Velocity errors are the quoted comparison-bin errors, with the separately documented figure-based estimate for NGC 4565. The central mass, radius and image coefficient are not retuned within their uncertainty ranges to obtain these entries. The input ranges, inclinations, projected axis ratios and radius status are recorded in Appendix E.

The objects are distinct disk galaxies represented by oblate surfaces. The calculation uses the outermost comparison radius jointly supported by the adopted kinematic and image data. Fourteen radii lie inside the last rotation point, while six are endpoints of their adopted datasets. UGC 7089 and UGC 8837 have no established catalogue flat speed. Table 3 therefore reports single-radius comparisons, with the radius status stated explicitly. The adopted global inventory should not be read as a verified enclosed baryonic mass at every interior radius.

The central absolute discrepancies range from 1.10% to 9.34%, with a median of 4.79%. The area pairs have Pearson r = 0.957 and the speed pairs r = 0.994. These describe the examples in Table 3. Equation (161) also shows that the area and speed residuals contain the same comparison information.

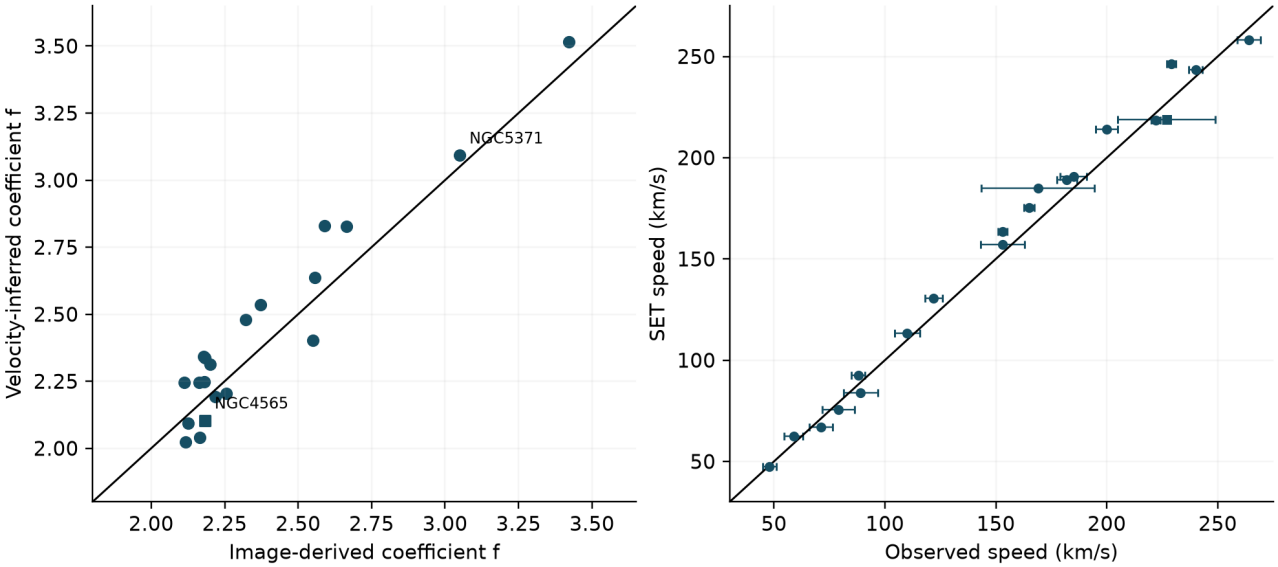


Figure 4. The twenty comparisons, with equality lines. Left: image-derived and velocity-inferred area coefficients. Right: predicted and observed speeds; horizontal bars show the comparison-bin velocity errors only. Squares identify NGC 4565, whose source provenance differs from the catalogue cases. The sample description and shared-input dependence are stated in the text.

8.2 NGC 5371

For NGC 5371, D = 39.7 Mpc gives a comparison radius R = 23.12 kpc. The adopted global baryonic inventory is approximately 1.851×10^{11} M⊙. The image axis ratio is about 0.7822; with i = 53 degrees it gives $\eta \approx 0.6256$ and $f_{image} \approx 3.049$. Using the retained full-precision inputs:

$$Q_{ref} = 1.67823 \times 10^{48}\ \mathrm{m^3/s}, \quad A_{image} = 4.87552 \times 10^{42}\ \mathrm{m^2}. \quad (162)$$

$$S_{boundary} = 344.215\ \mathrm{km/s}, \quad v_{SET} = 243.397\ \mathrm{km/s}. \quad (163)$$

The observed speed is 240.0 ± 3.0 km/s. The central difference is +1.42%, with $f_{inferred} \approx 3.092$. The larger oblate area lowers the thin-area prediction toward this bin. The comparison radius is half the last measured rotation radius of 46.24 kpc; its 240 km/s speed differs from the catalogue flat speed of 209.5 km/s. This is therefore an interior comparison using the stated boundary prescription. The inferred thickness depends on the oblate geometry and adopted inclination.

8.3 NGC 4565

For nearly edge-on NGC 4565, I use D = 9.7 Mpc and the last actual plotted HI point, at 420 arcsec, in the adopted Yim et al. curve [7]. This gives R ≈ 19.75 kpc. The S4G decomposition supplies the stellar luminosity, and the adopted HI flux is 274 Jy km/s [7, 8]. Equation (159) gives $M_{bar} \approx 6.540\times10^{10}$ M⊙. The

image ratio of about 0.2050 and inclination of 86.6 degrees give η ≈ 0.1965 and fimage ≈ 2.182. Keeping full precision in the calculation:

$$Q_{\mathrm{ref}} = 7.87756 \times 10^{47}\ \mathrm{m^3/s}, \quad A_{\mathrm{image}} = 2.54625 \times 10^{42}\ \mathrm{m^2}. \quad (164)$$

$$S_{\mathrm{boundary}} = 309.379\ \mathrm{km/s}, \quad v_{\mathrm{SET}} = 218.764\ \mathrm{km/s}. \quad (165)$$

The figure-derived observed speed is approximately 226.9 ± 22.0 km/s, giving −3.58%. The extraction leaves a small ambiguity about whether the plotted value already includes inclination correction; omitting that correction gives −3.41%. This remains an estimate from the published figure. The calculation record retains alternative distances as sensitivity checks, with mass and radius rescaled together.

8.4 The matched Newtonian baryonic comparison

To examine whether the area prescription improves the speed estimate, I compare it with the published Newtonian baryonic mass models at the same radii. These models calculate the radial force of the resolved gas, stellar disk and bulge [5]. They already include disk geometry. Their result is not the spherical reference √(GMbar/R).

The component forces add, giving the signed-quadrature expression:

$$v_N^2 = V_{\mathrm{gas}}\,|V_{\mathrm{gas}}| + 0.5V_{\mathrm{disk}}\,|V_{\mathrm{disk}}| + 0.7V_{\mathrm{bulge}}\,|V_{\mathrm{bulge}}|. \quad (166)$$

Here Vdisk and Vbulge are the published components at unit stellar mass-to-light ratio; the coefficients apply the same 0.5 and 0.7 convention as equation (159). Vgas already includes the adopted helium correction. The sign convention preserves any outward radial component. No dark-matter halo is added, and no component normalization is fitted to these velocities. The nineteen catalogue rows were checked against the published SPARC mass-model table at each exact adopted radius. NGC 4565 remains a supplementary SET point because a comparable Newtonian spatial model with the same adopted inventory has not been established here.

For example, NGC 4013 at R = 13.47 kpc has Vgas = 31.57, Vdisk = 154.56 and Vbulge = 70.18 km/s. The calculation gives vN = √[31.57² + 0.5(154.56²) + 0.7(70.18²)] = 128.02 km/s. At those same inputs the SET boundary prescription gives 190.73 km/s, compared with 185.0 ± 6.0 km/s observed. The uncertainty quoted here belongs to the measured speed, not to the complete prediction.

Figure 5 puts both calculations on common dimensionless axes. For each galaxy define vref = √(GMbar/R). The horizontal coordinate is the predicted speed divided by vref; the vertical coordinate is vobs/vref. Thus a SET point has horizontal coordinate 4/fimage, a Newtonian point has vN/vref, and equality with the observation lies on the diagonal. The dotted line at one identifies the spherical reference only. The SET estimate is calculated from Qref and Aimage; it is not obtained by multiplying the full Newtonian disk prediction by 4/fimage.

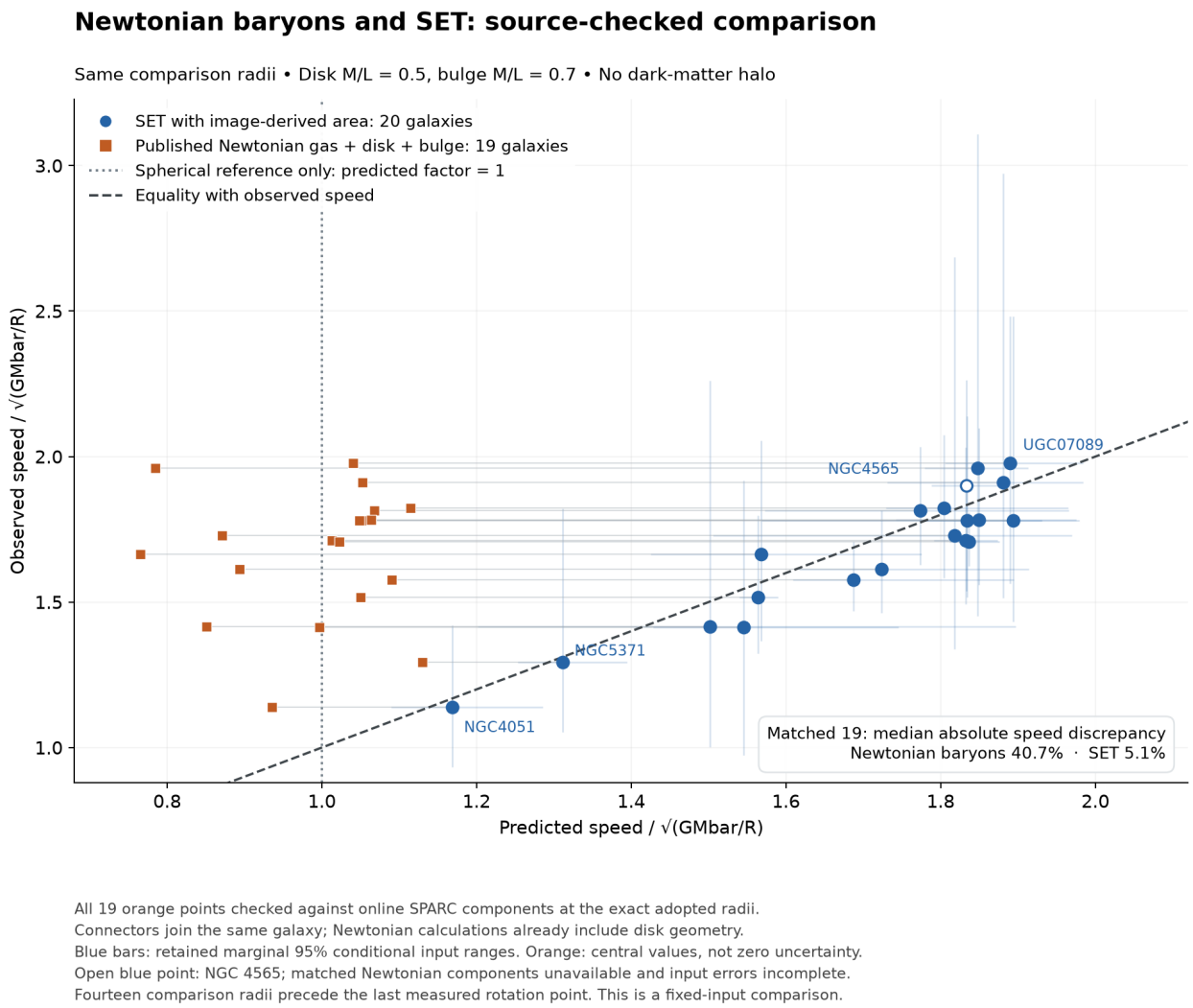


Figure 5. Published Newtonian baryonic components and the SET image-area prescription at the adopted comparison radii. Connectors join the same galaxy. Blue bars show marginal 2.5th–97.5th percentile ranges from joint input draws, with correlated axes; they are not complete prediction confidence intervals. Orange markers show central predictions, not zero uncertainty. The open blue point is NGC 4565. The matched-nineteen medians use $100|v_{pred}/v_{obs} - 1|$. Fourteen comparison radii are interior to the last measured rotation point. The source table already accounts for the Newtonian disk geometry.

Across the nineteen matched cases, the median absolute central speed discrepancy is 40.67% for the adopted Newtonian baryons and 5.13% for SET. The SET central discrepancy is smaller in each of these nineteen comparisons. The corresponding SET median for all twenty cases is 4.79%, as reported above. These are descriptive results for the cases in Table 3, with the same central inputs retained. The comparison concerns baryons at the stated stellar normalization, rather than a fitted dark-matter model or every possible baryonic inventory.

Here the image-area calculation improves agreement relative to spatially resolved Newtonian baryons, whose disk geometry is already included. The normalized plot displays the speed comparison on common axes. Appendix E gives the conditional input ranges, and the calculation record contains all nineteen component comparisons.

8.5 Source organization and measurement sensitivity

The proposal can now be stated physically. Mass establishes the source normalization. I then ask whether source organization changes the collective response in a way that can be expressed through an effective boundary area. At the same Mbar and R, a smaller effective area gives a larger boundary amplitude. The image-based implementation makes that idea calculable.

For each displayed case, the image coefficient comes from photometry and the adopted inclination. The central mass, radius and area are retained when calculating the speed. There is no separately adjusted halo mass or concentration in this calculation.

The diagnostic finferred asks which area the observed speed would require under this prescription. Comparing it with fimage tests the proposed geometric connection. Because both depend on some shared inputs, including inclination, their errors must be propagated together. The selection and correlation interpretation are stated with Table 3.

Figure 6 keeps the same masses, radii and observed speeds and changes only the area coefficient: a common f = 2 is replaced by each fimage. This shows how much the image correction changes these twenty predictions. The broader comparison with common normalizations is retained in the research supplement.

The differential dependence is:

$$d\ln v_{\mathrm{SET}} = \frac{1}{2} d\ln M_{\mathrm{bar}} - \frac{1}{2} d\ln R - d\ln f_{\mathrm{image}} . \quad (167)$$

Distance changes several inputs together. For a flux-derived inventory Mbar $\propto$ D^2, while a radius obtained from angular size has R $\propto$ D. Thus vSET $\propto$ $\sqrt{D}$ at fixed geometry. Inclination changes both the reconstructed area and the deprojected observed speed. Appendix E varies these inputs jointly and separates the stellar mass-to-light exercise. Its allowed-input envelopes show sensitivity; they are not confidence intervals or a simultaneous fit with shared systematic parameters.

The mass inventory and comparison boundary should describe the same system. Table 3 uses global inventories at the adopted comparison radii. Source reconstructions can determine how much material lies beyond each location and how that material contributes to the local burden and force. A matched mass aperture alone does not replace the field calculation for a general flattened system.

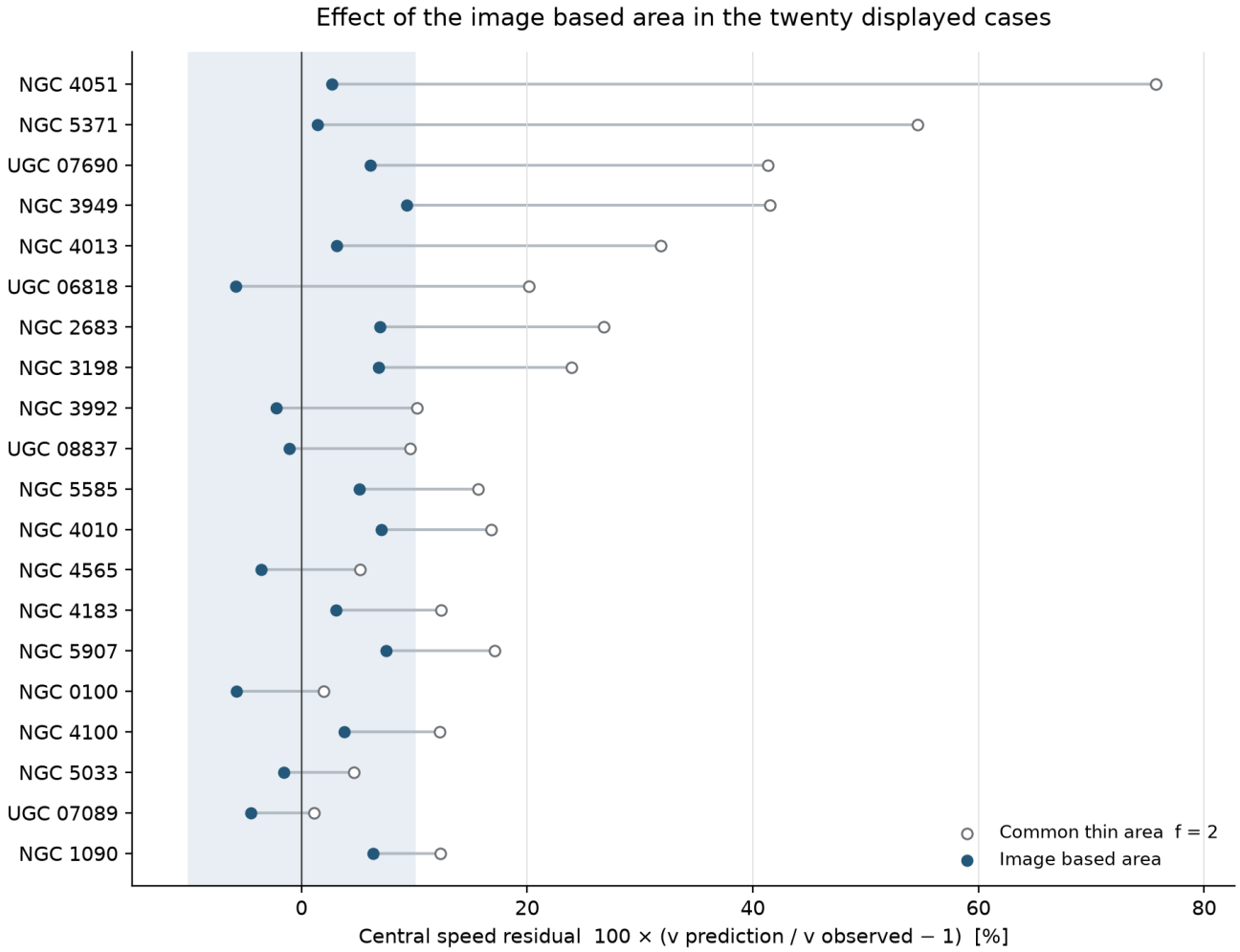


Figure 6. Signed speed residuals using f = 2 and the image-based f for the same twenty examples. Each line connects the two calculations for one galaxy. The shaded ±10% band is a descriptive reference, not a confidence interval.

8.6 Exterior continuation at farther measured radii

Table 3's comparison radii are observational evaluation locations. The local-field diagnostic in Section 7.3 tests the reduction at each location; measurements farther out separately test its proposed continuation.

As a separate consistency check, I retained the Table 3 masses, image coefficients and comparison radii and continued the fixed-shape exterior into the available outer velocity data. Fourteen galaxies have such measurements. At their last measured points, all fourteen central predictions are lower than the adopted observed speeds, with a median shortfall of 18.2%; two differ by less than 10%. The other six have no farther point in the adopted datasets. This checks that particular exterior continuation, rather than constituting twenty resolved-source field solutions. The tabulated velocity errors do not include a new complete propagation of source and geometric uncertainties. The full calculation record accompanies the single-radius comparisons.

9 Collective fields and energy accounting

The boundary reading and the stationary field describe connected but distinct parts of the calculation. I now put the source contribution into the invariant explicitly, retaining the specified response tensor and boundary condition. The same formulation supplies a check on the work associated with changes in source organization.

Appendix F states a stationary local-field formulation with a prescribed spatial response. Its equation is $-\nabla\cdot(C\nabla\Psi) = 4\pi G\rho\text{bar}$, with $\Psi = S^2/2$. The dimensionless symmetric positive tensor C describes the spatial response associated with that source organization. It is distinct from the remaining capacity $1 - u$ and the displacement metric q_{ij}.

Setting C to the identity recovers the ordinary weak-field source equation in burden variables.

The energy calculation identifies the work term associated with changes in the response.

9.1 The sources inside the invariant

We can now put the sources themselves into the invariant. For this stationary calculation I use Appendix F's spatially constant, symmetric positive-definite response tensor C and the boundary condition $S^2 \to 0$ at infinity. C specifies the response to source geometry; q_{ij} describes spatial displacement and α^2 the remaining capacity.

Let source a have mass M_a, reference radius R_a and volume $V_a = (4/3)\pi R_a^3$. Its nonnegative spatial profile $w_a(x')$ is normalized by $\int w_a(x')d^3x' = 1$, so $\rho\text{bar}(x') = \Sigma_a M_a w_a(x')$. The reference radius belongs to the source normalization; the observation position is x. Write $d = x - x'$. Substituting the stationary source solution into the capacity budget gives:

$$\frac{1}{12\pi\sqrt{\det C}}\sum\nolimits_a \{\frac{[4\pi\sqrt{2GM_aR_a^3}]^2}{(4/3)\pi R_a^3} \times \int \frac{w_a(\boldsymbol{x}')}{\sqrt{\boldsymbol{d}^T C^{-1}\boldsymbol{d}}}\, d^3x'\} + V_{\text{rel}}^2 + c^2\beta^2 = c^2 . \quad (168)$$

The first term is $S^2(x)$, written with its sources visible. The source distribution and specified response establish this field; the probe shares the remaining capacity between translation and event progression. A ship need not increase its intrinsic source normalization for a larger $Vrel^2$ contribution to leave less capacity for local events. The distinction is between source output and the probe's participation in the field. Here the ship is treated as a test probe in a prescribed field; its motion contribution is already represented by $Vrel^2$. All terms have units m^2/s^2.

The normalization simplifies directly:

$$\frac{[4\pi\sqrt{2GM_aR_a^3}]^2}{(4/3)\pi R_a^3} = 24\pi GM_a . \quad (169)$$

Using $\rho\text{bar} = \Sigma_a M_a w_a$ therefore gives:

$$S^2(\boldsymbol{x}) = \frac{2G}{\sqrt{\det C}}\int \frac{\rho_{\text{bar}}(\boldsymbol{x}')}{\sqrt{\boldsymbol{d}^T C^{-1}\boldsymbol{d}}}\, d^3x' . \quad (170)$$

The reference radius cancels from the normalized coefficient, while the source positions and density profiles remain in the integral. Subdividing the same density distribution into more numerical pieces leaves this field unchanged. For C = I and an isolated spherical exterior, equation (170) gives $S^2(r) = 2GM/r$. Equation (168) then reduces to:

$$V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2 - \frac{2GM}{r}. \quad (171)$$

This recovers the spherical amplitude as one solution for S. For the collective boundary proposal, I retain $Q\Sigma = Q_{\mathrm{ref}}$ and $A_{\mathrm{eff}} = f\pi R^2$. Where the boundary amplitude equals the local field, $S = S_{\mathrm{eff}} = (4/f)\sqrt{(2GM/R)}$. The fixed-Q identity in equation (147) follows at the same M and reference R. Here $f \to 2$ is the formal thin-area limit.

Writing the source term inside the invariant retains the specified response and boundary conditions. In this static timelike description $S^2 + V_{\mathrm{rel}}^2 < c^2$. Sections 3 and 4 give the spherical action, clock and propagation relations.

9.2 Response and mechanical energy

For $\Psi = S^2/2$, the stationary source equation is $-\nabla\cdot(C\nabla\Psi) = 4\pi G\rho_{\mathrm{bar}}$. Appendix F gives its integral solution and the boundary orientation argument. For finite smooth sources, the binding energy is $W = -\tfrac{1}{2}\int\rho\Psi\, d^3x$. When the prescribed response changes, the energy balance is

$$\frac{d(E_{\mathrm{kin}} + W)}{dT} = \frac{1}{8\pi G}\int \nabla\Psi \cdot \frac{\partial C}{\partial T}\nabla\Psi\, d^3x. \quad (172)$$

A fixed response conserves this mechanical energy. A changing response supplies or removes work through the displayed term. If the response depends on source coordinates, its variation also enters their force. Appendix F retains the full derivation, including that source-dependent contribution. This accounting makes the physical meaning of a change in collective response explicit.

10 Cosmological source evolution and physical clocks

The cosmological question follows the same source–capacity reasoning in a different setting. Instead of reading a stationary exterior at successive radii, I follow an evolving material domain and the collective region included in its burden. The observable will be the brightness received at a measured redshift. The volume evolution and front prescription are stated before that brightness is calculated.

10.1 Source volume and receiving clock

Space Emanation Theory proposes that matter establishes a burden in the field through which physical interactions occur. Gravitational burden, motion and the progression of local events participate in one finite capacity. A clock measures that progression through its physical ticks. We examine a homogeneous cosmological realization of these principles and calculate how its source and clock relations affect the brightness received from distant supernovae.

The source law and the local capacity relation are

$$Q^2 = 24\pi GMV, \quad S^2 + V_{\mathsf{rel}}^2 + c^2\beta^2 = c^2 . \quad (173)$$

Here M is the baryonic mass assigned to the source domain and V is its volume. The law specifies the positive source quantity Q, with units $\mathsf{m}^3\,\mathsf{s}^{-1}$, using reference time T. The gravitational burden S is expressed in speed units; S^2 is its contribution to the capacity budget. V_{rel} describes local motion relative to the field. The dimensionless factor β describes the progression of a physical clock relative to T, and c sets the capacity scale. Volume V and velocity V_{rel} are distinct quantities.

For a spherical domain, $V = 4\pi R^3/3$ and $M = \rho V$. The source law gives

$$Q = 4\pi\sqrt{2GMR^3}, \quad H_b \equiv \frac{Q}{3V} = \sqrt{\frac{8\pi G\rho}{3}} . \quad (174)$$

The second equality is direct algebra. To use this rate for physical evolution, we adopt $dV_m/dT = Q_m$ for a material domain containing a fixed collection of baryonic matter. The subscript m distinguishes this domain from the advancing collective region introduced in Section 10.2. Its volume scales as $V_m \propto a^3$, where a is the common scale factor and $a = 1$ at reception. Conserved material mass then gives $\rho = \rho_0 a^{-3}$.

$$H_T \equiv \frac{1}{a}\frac{da}{dT} = \frac{Q_m}{3V_m} = H_b(a) = H_{b0}a^{-3/2} . \quad (175)$$

The factor three comes from differentiating a volume proportional to a^3. At any given epoch, $Q/(3V)$ has the same value for domains with the same uniform density. A fixed material mass does not require a constant Q_m: the source law also contains the evolving volume. Present-day values carry the subscript zero.

The same process read by a physical clock

The time T is the reference used to express the source law. The variable t records the reading of the comoving physical clock used in this cosmological calculation. SET interprets its ticks as events realized within the available field capacity. A smaller clock factor means fewer ticks per reference second. The relation between these two time readings is part of the physical description being tested here.

A comoving clock follows the local homogeneous matter and has $V_{\mathsf{rel}} = 0$ in this construction. The capacity relation therefore gives $\beta = \alpha$ and

$$u \equiv \frac{S^2}{c^2}, \quad \alpha = \sqrt{1-u}, \quad dt = \alpha\, dT. \quad (176)$$

For $0 \le u < 1$, the clock factor is positive. The same source-driven volume increment can be expressed using either time reading:

$$dV_m = Q_m\, dT = Q_{\text{normalized},m}\, dt. \quad (177)$$

Dividing by the physical clock interval gives the clock-normalized source rate,

$$Q_{\text{normalized},m} = Q_m \frac{dT}{dt} = \frac{Q_m}{\alpha}. \quad (178)$$

The source law is expressed per reference time. A clock within the burdened field registers a smaller interval over the same process, so expressing that process per second of this clock gives a larger numerical rate. The amount assigned to a given physical interval is unchanged. For example, if $\alpha = 1/2$ throughout an interval, one reference second corresponds to half a clock second and the clock-normalized rate is $2Q_m$. The factor follows from the clock conversion; it is not an extra source contribution.

We use this reading to calculate the evolution measured by the comoving clock:

$$H \equiv \frac{1}{a}\frac{da}{dt} = \frac{1}{3V_m}\frac{dV_m}{dt} = \frac{Q_{\text{normalized},m}}{3V_m} = \frac{H_b}{\alpha}. \quad (179)$$

For any selected domain in this homogeneous prescription, we write $Q_{\text{normalized}} = Q/\alpha$; the domain label can be omitted when it is clear. As the collective burden changes, α changes, and the source rate is read differently by physical clocks. Section 10.2 calculates this history. Section 11.1 uses it in the distance and brightness calculation. The inverse clock factor is already included in $H(z)$, so it is not applied again to the final brightness.

10.2 The collective burden history

To calculate α through time, we specify the collective source region that supplies the burden. This test uses uniform density inside a moving spherical front of radius R_h, with $M_h = 4\pi\rho R_h^3/3$. The radius identifies the domain included in this burden readout. Its kinematic evolution, expressed in the same clock time t, is

$$\frac{dR_h}{dt} = HR_h + c. \quad (180)$$

The first term carries the front with the evolving scale; the second advances it locally at c. We adopt this evolution for the collective boundary. Its enclosed mass can change as it advances, whereas the material domain used to define the source-volume rate has fixed mass.

For the burden readout we retain SET's isotropic static source kernel: a mass element dm at distance r contributes $dS^2 = 2G\, dm/r$. At the centre of a uniform sphere, a shell of thickness dr contains $dm = 4\pi\rho r^2 dr$. Adding these shell contributions gives

$$S_c^2 = 2G\int_0^{R_h} \frac{\rho}{r}\, 4\pi r^2\, dr = 4\pi G\rho R_h^2 = \frac{3GM_h}{R_h}. \quad (181)$$

This is the central burden of the collective sphere. The spherical exterior expression $S^2 = 2GM/r$ describes a different evaluation location; it is not substituted for this interior sum. Since $H_b^2 = 8\pi G\rho/3$, the central result becomes

$$u = \xi \frac{H_b^2 R_h^2}{c^2}, \quad \xi = \frac{3}{2}. \quad (182)$$

The factor $3/2$ is the central uniform-source factor, fixed by this readout rather than adjusted numerically. We apply the static uniform-source kernel to the evolving domain as the burden prescription for this calculation. Together with the front evolution and source-volume law, it specifies the physical realization being tested.

Differentiate the burden expression with respect to $\ln a$. The density contributes -3, while the advancing radius contributes $2\, d\ln R_h / d\ln a$. Using the front and clock relations gives

$$\frac{du}{d\ln a} = u\left[-3 + 2\left(1 + \frac{c}{HR_h}\right)\right] = -u + 2\sqrt{\xi u(1-u)}. \quad (183)$$

The substitution $y = \sqrt{u/(1-u)}$ makes the numerical evolution especially simple:

$$\frac{dy}{d\ln a} = \frac{1+y^2}{2}\left(2\sqrt{\xi} - y\right), \quad H = H_{b0} a^{-3/2}\sqrt{1+y^2}. \quad (184)$$

This is the response equation used for the brightness calculation. The source law supplies the density-dependent rate; capacity supplies the clock conversion; the stated uniform-domain and front prescriptions supply the burden history. Once those prescriptions are retained, the differential equation follows without an adjustable dimming exponent.

11 Supernova brightness and observations

11.1 From the source history to received light

For the photon calculation we retain the homogeneous, spatially flat propagation coupling, with photon number conserved. It relates emission and reception by $1 + z = a_o/a_e$, stretches clock intervals by $dt_o/dt_e = 1 + z$, and reduces each photon's received energy by the same factor. With $a_o = 1$, the luminosity distance and bolometric flux are

$$D_L(z) = (1+z)c\int_0^z \frac{dz'}{H(z')}, \quad F_o = \frac{L_e}{4\pi D_L^2}. \quad (185)$$

In catalogue calculations, the integral uses z_{HD} and the prefactor uses $1 + z_{\mathsf{HEL}}$, following the published redshift convention. The released brightnesses already include light-curve standardization and observational corrections [15, 16, 17, 18]. The test compares these standardized brightnesses across different supernovae, rather than the fading of one explosion over several days.

Define $E(z) = H(z)/H_0$ and a dimensionless luminosity distance $d_L = H_0 D_L/c$. The tested history gives

$$E(z) = (1+z)^{3/2}\sqrt{\frac{1+y(z)^2}{1+y_0^2}}, \quad d_L = (1+z_{\mathsf{HEL}})\int_0^{z_{\mathsf{HD}}} \frac{dz'}{E(z')}. \quad (186)$$

The initial source-volume calculation has a constant clock factor, so its normalized history is $E_0(z) = (1+z)^{3/2}$. Its distance is

$$d_{L,0} = 2(1+z_{\mathsf{HEL}})\left[1 - (1+z_{\mathsf{HD}})^{-1/2}\right]. \quad (187)$$

This is the matter-only, spatially flat baseline with a constant clock factor. Its normalized distance shape is also the classical dust cosmology shape. The comparison asks whether the evolving SET source–capacity prescription accounts for the observed brightness departure from this baseline while the source inputs remain fixed. Agreement is assessed against catalogue-standardized brightness and redshift; the physical interpretation is considered separately. The plotted baseline is not a fitted ΛCDM model.

For each target group, let $\mathscr{L}_i[x]$ mean its weighted mean of x minus the weighted mean in the fixed nearby reference group, $0.025 < z_{\mathsf{HD}} < 0.10$. Its weights sum to zero. Therefore an unknown constant luminosity or distance zero point cancels. If m denotes the released standardized magnitude or distance modulus, the plotted quantities are

$$B_{\mathsf{obs},i} = 10^{-0.4\mathscr{L}_i[m - 5\log_{10} d_{L,0}]}, \quad (188)$$

$$B_{\mathsf{SET},i} = 10^{-0.4\mathscr{L}_i[5\log_{10}(d_L/d_{L,0})]}. \quad (189)$$

The initial prediction is $B_0 = 1$. A value below one means dimmer than the low-redshift-anchored initial curve. The predicted contrast uses the catalogue redshifts and the fixed source calculation; observed brightness does not enter its numerator. Every object is evaluated at its own redshift before averaging. These are geometric-mean relative fluxes, because averaging is performed in magnitudes.

11.2 Why the brightness departure motivated dark energy

Type Ia supernovae allow us to compare received brightness with the brightness predicted at a measured redshift. Their light curves are standardized to account for known luminosity differences, and nearby supernovae establish a common reference. The question is then whether the predicted distance–redshift relation accounts for the brightness of the more distant sample.

The original supernova studies found distant events to be systematically fainter than expected in the matter-only cosmologies they examined. With standardized luminosity L, the flux relation $F = L/(4\pi D_L^2)$ assigns a larger luminosity distance to a fainter source. Fits to the full distance–redshift relation favored histories with recent accelerated expansion, which a positive cosmological constant can produce in GR cosmology. Acceleration was a quantitative inference from the brightness data under those models, with observational systematics assessed [21, 22]. SET investigates whether its evolving collective burden, clock relation and source-driven history can account for the brightness departure without inserting an additional dark-energy density.

The modern comparison below uses a specified matter-only, spatially flat baseline. It illustrates the brightness departure with the present catalogues; it is not a reconstruction of the original discovery analysis. For the Pantheon+ group near $z = 0.731$, the baseline predicts relative brightness $B_0 = 1$, while the observed value is $B_{\mathrm{obs}} = 0.700656$. The observation is therefore 29.93% dimmer than this baseline after the nearby-reference normalization.

The SET history predicts $B_{\mathrm{SET}} = 0.706666$, or 29.33% dimmer than the same baseline. The corresponding effective relative distance factors are

$$B_{\mathrm{obs}}^{-1/2} = 1.1947, \quad B_{\mathrm{SET}}^{-1/2} = 1.1896\,. \quad (190)$$

These are group contrasts relative to the nearby reference, not absolute distances to a single supernova. The baseline predicts too much brightness, and the SET calculation reduces it by almost the amount required. Section 11.3 shows the source inputs and numerical steps. Its 42.72% and 0.86% discrepancies use observed brightness as the denominator; the dimming percentages above instead use the baseline.

The brightness departure that motivated dark-energy cosmology is also the observational target of this SET calculation. Here, the evolving collective burden and its clock conversion change the predicted distance–redshift relation without inserting an additional dark-energy density. The same prescription moves the predictions toward observation in all fifteen redshift groups. Across the full grouped comparison, the median absolute discrepancy falls from 38.53% to 2.09% in Pantheon+ and from 36.39% to 2.96% in DES.

11.3 One complete calculation

Measured inputs and the source calculation

Use the Pantheon+ group $0.65 \le z < 0.80$: 63 supernovae with weighted mean redshift 0.73085168, compared with the fixed 311-object nearby reference. This is the closest central group match in the run; the full sample follows in Figure 7 and Table 6. The present-day source inputs are

$$M_0 = 1.5005279083 \times 10^{53}\ \mathrm{kg}, \quad R_0 = 4.3992396698 \times 10^{26}\ \mathrm{m}\,. \quad (191)$$

The radius is 46.5 billion light-years. These frozen benchmark inputs use a cosmology-conditioned radius and imply

$$V_0 = 3.5663295987 \times 10^{80}\ \mathrm{m^3}, \quad Q_0 = 5.1893886387 \times 10^{62}\ \mathrm{m^3\,s^{-1}}, \quad (192)$$

$$H_{b0} = 14.96662955\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \quad u_0 = \frac{3GM_0}{R_0 c^2} = 0.7598922704\,. \quad (193)$$

Thus $\alpha_0 = 0.4900078873$, $y_0 = 1.7789876897$, and $H_0 = 30.54365029\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. The initial condition is fixed before calculating the brightness contrast.

Expected matter only baseline

Evaluate the initial distance expression at every target and reference redshift. After the common reference normalization, this baseline has $\Delta m_0 = 0$ and expected relative brightness $B_0 = 1$. The value one is a defined comparison scale, not an absolute flux measurement.

Expected SET calculation

Integrate the burden history from the fixed initial condition, calculate the distances, and average the magnitude differences using the same weights. This gives

$$\Delta m_{\mathrm{SET}} = \mathscr{L}[5\log_{10}(d_L/d_{L,0})] = 0.37696377\ \mathrm{mag}, \quad (194)$$

$$B_{\mathrm{SET}} = 10^{-0.4(0.37696377)} = 0.70666648\,. \quad (195)$$

Observed brightness and comparison

The catalogue gives $\Delta m_{\mathrm{obs}} = 0.38623777 \pm 0.03403476\ \mathrm{mag}$, equivalent to $B_{\mathrm{obs}} = 0.70065608$ with marginal 1σ interval 0.67903317–0.72296754. Define the signed discrepancy as $100(B_{\mathrm{pred}}/B_{\mathrm{obs}} - 1)$.

Table 4. Initial prediction SET prediction and observed brightness

Calculation	Relative brightness	Discrepancy
Initial matter-only baseline	1.00000000	+42.72%
SET source-clock calculation	0.70666648	+0.86%
Observed group	0.70065608	—

The remaining difference is 0.00927399 mag, or 0.27 marginal standard deviations. The SET calculation reduces a 42.72% central discrepancy to 0.86%, placing the prediction comfortably inside the observed 1σ interval.

11.4 The same calculation across both catalogues

Figure 7 shows the unchanged calculation for every fixed group. No group is removed for disagreeing. The orange line connects the group predictions; it is not a curve fitted through the observations.

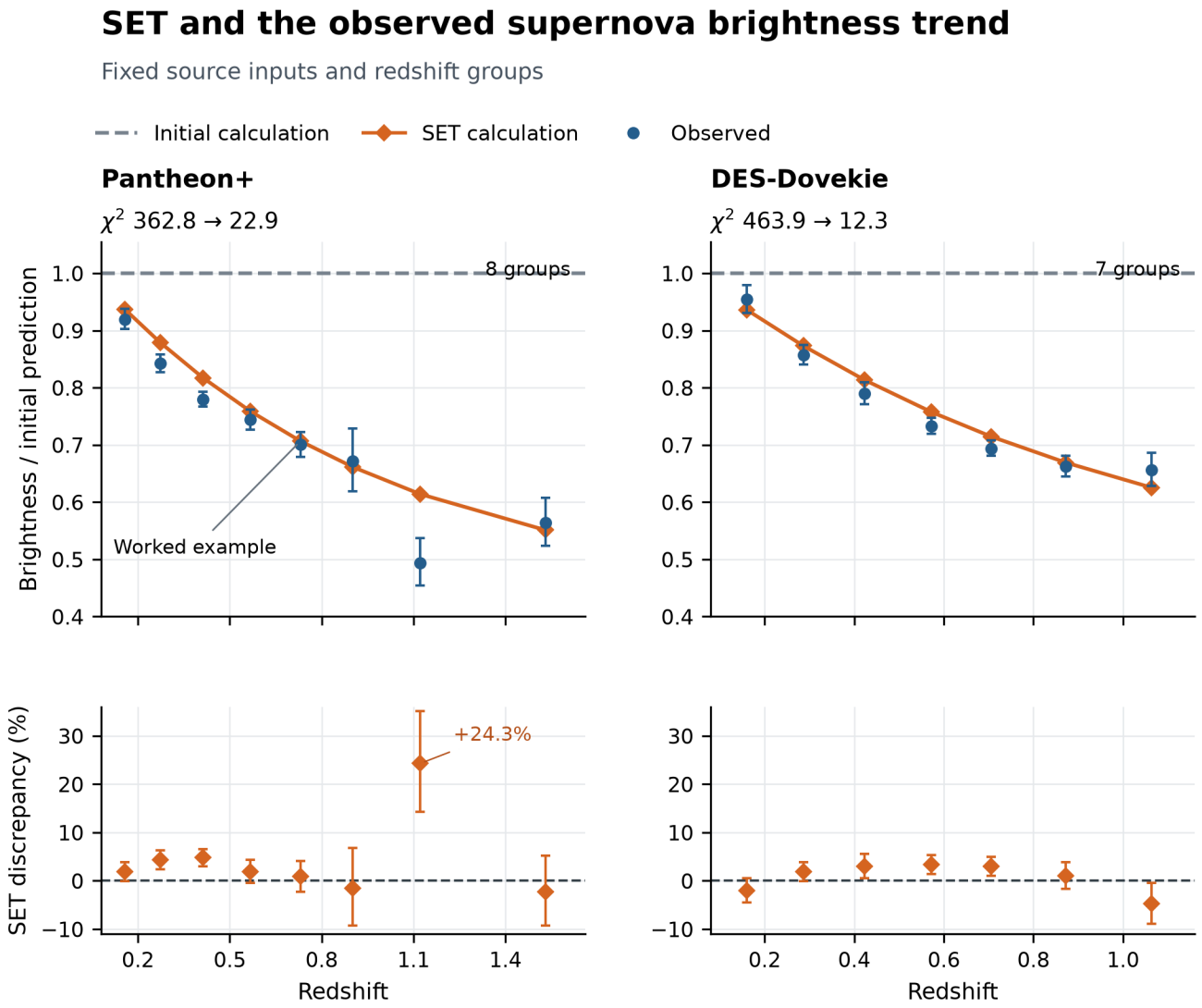


Figure 7. Relative supernova brightness and remaining SET discrepancy. Blue bars are marginal 1σ observational intervals. The initial source-volume prediction is one after reference normalization. The lower panels retain the sign of the remaining discrepancy. Formal comparisons use the full correlated magnitude covariance, including the common nearby reference. Source inputs are held fixed.

Table 5. Grouped comparison statistics with the fixed source prescription

Comparison	Pantheon+	DES-Dovekie
Initial χ^2	362.75	463.91
SET χ^2	22.87	12.34
Median absolute error before	38.53%	36.39%
Median absolute error after	2.09%	2.96%

All fifteen central discrepancies decrease in magnitude. Eight group predictions lie within their marginal 1σ intervals: five of eight in Pantheon+ and three of seven in DES. These counts describe grouped means, not the fraction of individual supernovae explained. The two catalogue statistics must not be added as independent evidence.

11.5 Complete grouped results

Table 6 retains all plotted values. N is the number of distinct catalogue object identifiers in the target group; $\bar{z}$ is its weighted mean redshift. The observed interval is marginal 1σ. The initial prediction is one in every row. Before and after are signed percentage discrepancies relative to the observed central brightness.

Data	$\bar{z}$	N	Observed and 1σ interval	SET	Before	After
P	0.1575	207	0.919799 [0.902394, 0.937540]	0.936955	+8.72%	+1.87%
P	0.2729	380	0.842757 [0.827165, 0.858642]	0.879009	+18.66%	+4.30%
P	0.4131	163	0.779774 [0.766711, 0.793059]	0.817031	+28.24%	+4.78%
P	0.5675	117	0.744370 [0.726833, 0.762331]	0.758733	+34.34%	+1.93%
P	0.7309	63	0.700656 [0.679033, 0.722968]	0.706666	+42.72%	+0.86%
P	0.9018	5	0.671954 [0.619381, 0.728991]	0.661347	+48.82%	-1.58%
P	1.1217	9	0.494130 [0.454592, 0.537105]	0.614079	+102.38%	+24.27%
P	1.5302	16	0.563875 [0.523682, 0.607152]	0.551170	+77.34%	-2.25%
D	0.1608	44	0.954695 [0.930655, 0.979356]	0.935815	+4.75%	-1.98%
D	0.2863	309	0.857211 [0.840490, 0.874264]	0.873224	+16.66%	+1.87%
D	0.4225	404	0.789908 [0.771064, 0.809211]	0.813755	+26.60%	+3.02%
D	0.5726	467	0.733203 [0.719298, 0.747377]	0.757433	+36.39%	+3.30%
D	0.7059	265	0.694030 [0.680790, 0.707528]	0.714577	+44.09%	+2.96%
D	0.8721	113	0.662420 [0.644568, 0.680767]	0.669145	+50.96%	+1.02%
D	1.0639	17	0.656663 [0.628014, 0.686618]	0.625400	+52.29%	-4.76%

Table 6. Full results of the retained central uniform-burden calculation. P denotes Pantheon+ and D denotes DES-Dovekie. Values are rounded for presentation; calculations use the full stored precision.

The target intervals are [0.10, 0.20), [0.20, 0.35), [0.35, 0.50), [0.50, 0.65), [0.65, 0.80), [0.80, 1.00), [1.00, 1.30), and [1.30, 2.40). DES has no object in the final interval. The nearby references contain 311 Pantheon+ objects and 201 DES objects. Pantheon+ rows below $z = 0.025$ are not used in this fixed-reference plot.

Within each group and reference, weights use the published marginal inverse variance, with duplicate rows sharing object weight. The full statistical and systematic covariance is then propagated. This weighting was adopted in a documented amendment after an equal-object calculation exposed extremely uncertain DES entries; the redshift boundaries were unchanged. No weight or inclusion rule depends on a SET residual.

The sparse Pantheon+ group near $z = 1.122$ remains a visible exception to close central agreement: the discrepancy falls from +102.38% to +24.27%. Its nine objects remain in the table and figure.

11.6 Covariance and verification

The plotted measurements come from released standardized supernova products, not raw photon counts. Pantheon+ uses the published SN-only cut $z_{\mathrm{HD}} > 0.01$ before the fixed reference and group definitions are applied. DES uses the 1,820-row Dovekie release. The plotted target and reference groups contain 1,271 distinct Pantheon+ identifiers and all 1,820 DES identifiers, with overlap between catalogues.

For the formal statistic, stack the weighted contrasts into a matrix $\mathscr{L}$. If C is the released magnitude covariance and r the residual vector, the calculation is

$$C_{\mathrm{contrast}} = \mathscr{L} C \mathscr{L}^{\mathsf{T}}, \quad \chi^2 = r^{\mathsf{T}} C_{\mathrm{contrast}}^{-1} r . \quad (196)$$

The DES release supplies an inverse covariance, which is solved against to form the contrast covariance. It is not treated as the covariance itself. Error bars in Figure 7 are the marginal diagonal uncertainties, whereas this statistic includes correlations. The uncertainty bands are conditional on the frozen SET inputs; they are not model-parameter uncertainty bands.

A separate unbinned check retained the full released covariance and analytically profiled one constant luminosity intercept. Its initial and SET statistics were 2101.15 and 1426.45 for Pantheon+, and 2558.52 and 1647.64 for DES. This is a different check from the fixed-reference figure: it fits one nuisance zero point, but no SET parameter. It supports a broad improvement without establishing that every object improves.

The companion verification script recomputes all fifteen predictions directly from the fixed history and the catalogue redshifts, then checks them against the archived table. The forward predictions agree to better than 10^{-12} in relative brightness. The source history has also been cross-checked against an analytic integral of its evolution equation. The source records, plot and input hashes are retained with this section.

The research compared several physical prescriptions, including central and boundary burden readouts, before retaining this expression. Source parameters were held fixed, but this was exploratory development rather than a blind prediction. The reported χ^2 values describe these comparisons and are not corrected for the search across candidate histories.

The comparison uses relative brightness, for which the common absolute distance scale cancels. It therefore tests the redshift dependence of the prediction without independently validating its absolute scale. Additional distance–redshift and domain checks for this prescription are documented in Appendix J.

The original research also checked exact catalogue overlap. Removing the 339 shared identifiers from both catalogues, while retaining the same response and redshift intervals, leaves a reduction in the correlated brightness statistic from 265.77 to 18.54 in Pantheon+ and from 187.09 to 8.82 in DES. The latter reference then contains only ten objects, and common calibration remains. This supports persistence of the improvement, but is not a blind independent replication.

12 Physical interpretation and results

I began with clocks because every clock reading is itself a physical event. The capacity relation gives a common language for that event progression, motion and gravitational burden. The source law then connects the burden calculation to matter. What matters is following that connection all the way to a quantity that can be compared with observation.

12.1 The spherical construction

The field calculation requires the physical work-current to carry the source charge and to agree with the source-surface law across a universal family. This selects both $S^2 = 2GM/r$ and $\zeta^2 = 1/A$ on the family's covered interval. Stationary calibrated event progression supplies the probe action, with its coefficient fixed by inertial mass. That action gives the conserved orbital energy and the exact orbit. The finite and differential current identities show explicitly how the original burden and its held-clock normalization describe the same gradient. The normalized-burden orbital equation preserves the same motion through the complete change of trajectory parameter.

The logical dependencies are explicit:

Table 7. Principles and derived relations in the spherical calculation

Quantity or relation	Status in this construction
Capacity relation	Retained SET capacity principle
Probe action	Stationary event progression with coefficient fixed by inertial mass
Source law and source-surface correspondence	Retained source principles
Physical-current balance and universal $\mathcal{H}(\mathrm{A})$	Specified spherical field-response premises
Radial conversion and exterior burden	Determined by consistency and the exterior boundary condition
Finite and differential current normalization	Derived algebraically from capacity and the specified work-current measure
Normalized orbital potential	Derived by an exact change of orbital parameter
Perihelion equation and leading advance	Derived from the selected field and probe action

The solved clock and motion relations are mathematically equivalent to the Schwarzschild exterior. That equivalence is a result to check; an Einstein field equation or a metric-completion action did not select the response in this route. Appendix D obtains the familiar free-fall representation after the field has been solved.

Orbital energy and field energy answer different questions. Sections 3.3 and 5.1 derive the probe's conserved energy and its normalized-burden representation. A field-energy functional assigns energy to field configurations; an appropriate variational principle can then supply field equations. Here the stationary equations are stated directly through current balance and the response class. Selecting a separate field functional is not required to solve this exterior problem. Equivalent action expressions can yield the same equations; the relevant requirement is determinate dynamics.

With the physical-current balance, universal response class and source-surface relation stated above, the spherical construction determines the exterior burden and radial response over the covered interval. The probe action then gives the conserved orbital energy and perihelion equation. The normalized-burden formulation preserves the same motion through an exact change of orbital parameter.

12.2 Collective sources and received light

For collective systems, I propose that source arrangement enters through an effective boundary area. The twenty examples show how this prescription connects adopted baryonic mass and image geometry to observed speeds. Within the nineteen matched comparisons it improves agreement over the adopted Newtonian baryonic calculations. The sample and uncertainty conditions accompanying those results determine what can be inferred from them.

Appendix F supplies a stationary source equation for a prescribed response and the work associated with changes in that response. The source-boundary algebra, the galaxy comparisons and this field formulation are the results presented here. I invite independent reproduction using the stated inputs and calculation record.

SET's physical proposal is that the baryonic sources participate in the same field capacity through which clocks, motion and light are realized. In this calculation, the source law fixes a rate per reference time. The collective burden determines how that reference rate is read by physical clocks. The moving front changes the burden history, and the history changes the integrated distance associated with a measured redshift. Photon energy, arrival rate and geometric spreading then determine the brightness.

The specific realization tested here includes a changing physical scale a. The brightness change comes from the coupled history of source-driven motion, clock progression and light propagation. No additional dark-energy density is inserted into these equations. A static delay along the same path would not by itself give this result: the operative quantity here is the evolving distance–redshift relation.

This connects to SET's broader motivation. The galaxy calculation investigates whether baryonic organization can alter the gravitational response through collective burden and effective area. Here the question concerns the history of a collective response and the light received through it. The source–capacity connection motivates both investigations, while the specific mathematical assumptions and observations differ.

The empirical value is the breadth of the improvement. A single fixed expression reduces the discrepancy in every group across two published compilations. The data were included by redshift and published quality information, rather than agreement with SET. The result concerns a trend supported by many observations, rather than one fortunate match. The median absolute central discrepancy falls from 38.53% to 2.09% in Pantheon+ and from 36.39% to 2.96% in DES.

The result presented here is that one fixed SET prescription moves the predicted relative brightness toward observation in all fifteen redshift groups. Its physical significance is the demonstrated connection between the proposed source–capacity mechanism and the observed brightness trend. Whether this mechanism changes the inferred cosmic expansion history must be assessed through its predictions for observable quantities.

The connection I am investigating is that the organization of baryonic sources and the history of their collective burden can affect what we infer from motion and light. In the galaxy application, the effective area enters a proposed boundary response. In the cosmological application, the source law, clock conversion and evolving collective region enter a distance–redshift history. The observables differ, and their specific response prescriptions are displayed, but the physical question is shared.

The calculations give that question numerical form: a selected spherical exterior and its perihelion motion, a rotating strong-field construction, an image-area comparison with galaxy speeds, and a source-clock calculation of supernova brightness. I invite readers to reproduce these steps, examine the physical correspondences and investigate whether the source–capacity account can be developed across these scales.

12.3 Data and calculation records

The research supplement accompanying Draft 48, SET-research-supplement.md, preserves the catalogue context and comparison controls. The source and calculation bundle contains the twenty central rows, sensitivity ranges, observational provenance, archived input files, figure data and the scripts used to build

and check this manuscript. It also includes the nineteen matched Newtonian component calculations and the conditional input-draw record used in Figure 5. The calculation record also includes the boundary-applicability audit, distinguishing the prescribed exterior continuations from the separately reconstructed local field of NGC 3198. Appendices I and E give the numerical-check summary and the inputs required to trace the displayed examples. These materials are part of the research record; a public archival identifier can be assigned with submission.

The numerical checks accompanying this derivation include exact algebraic identities, direct orbit integrations, independent angular quadratures and the finite clock integrals in Appendix C. They validate the displayed calculations under their stated premises. They do not constitute additional astronomical measurements.

The derivation, galaxy and cosmological records remain available with their original manuscripts. The combined numbering used here is accompanied by an equation and section correspondence record. The numerical tables and source figures are retained from the stated calculations; combining them in one paper does not make their datasets independent. The primary catalogues and observational sources are listed in the references.

Appendix A Source family selection controls

Write $x = 1 - A_R$ and define the integrated response

$$\mathscr{F}(x) = \int_0^x \frac{dt}{\sqrt{\mathscr{H}(1-t)}}. \quad (197)$$

The source-field correspondence requires $\mathscr{F}(x) = x$. At only one chosen boundary $x = x_0$, this is one integral constraint. As an explicit control, take

$$\mathscr{F}(x) = x + \eta x^2(x - x_0), \quad x_0 = \frac{1}{2}, \quad \eta = \frac{1}{5}. \quad (198)$$

$$\mathscr{F}'(x) = 1 + \eta(3x^2 - 2x_0 x), \quad \mathscr{H}(1-x) = \frac{1}{[\mathscr{F}'(x)]^2}. \quad (199)$$

This derivative is positive on $0 \le x \le 1$. The response has $\mathscr{H}(1) = 1$ and satisfies $\mathscr{F}(x_0) = x_0$, yet

$$\mathscr{H}(1 - x_0) = \frac{1}{(1 + \eta x_0^2)^2} = 0.90702948 \neq 1. \quad (200)$$

This is a mathematical control, not a proposed complete alternative theory. It shows why one boundary agreement does not select a function. Requiring $\mathscr{F}(x) = x$ on a continuous interval instead gives $\mathscr{F}'(x) = 1$ and hence $\mathscr{H} = 1$ on that interval.

A second control follows an alternative through the actual field equation. Let $\mathscr{H}(A) = 1 - b(1-A)^2$, with $b = 1/2$, and put $y = R_s/r$ and $x = 1 - A(r)$. The physical-current equation gives

$$\frac{dx}{dy} = \sqrt{1 - bx^2}, \quad x(0) = 0, \quad x(y) = \frac{\sin(\sqrt{b}\, y)}{\sqrt{b}}. \quad (201)$$

At a source surface, the source law requires $x(y_R) = y_R$. The displayed alternative instead has $x < y$ for the relevant positive compactness values. It cannot retain both the same current law and the same continuous source-surface correspondence. If one keeps $x = y$ instead, its current is $r^2\mathscr{J} = GM/\sqrt{\mathscr{H}}$, which violates the specified current balance. The selection therefore uses the combined field and source premises, before any orbit is compared.

Appendix B Alternative radial response and exact orbits

To isolate what normalization does, hold the same exterior profile $A = 1 - R_s/r$ but choose the flat-radial control $\zeta^2 = 1$. This control shares the held readout $Q_\star$ with the selected solution. It does not satisfy all the field-selection premises of Sections 3.3 and 3.4.

The exact auxiliary equation from Section 5.1 becomes

$$r_{\sigma\sigma} - r\varphi_\sigma^2 = -\frac{GM}{(r - R_s)^2}, \quad r^2\varphi_\sigma = \ell_\sigma . \quad (202)$$

$$w'' + w = \frac{GM}{\ell_\sigma^2(1 - R_s w)^2} . \quad (203)$$

This is exactly the spatial orbit of the earlier shifted-potential calculation, provided its angular momentum is identified as $\ell_\sigma = j/\varepsilon$ and its time is interpreted as σ. The match holds beyond the leading perihelion coefficient. Calling that auxiliary force a force in reference time T would be a different dynamical prescription.

For a direct comparison, choose the same two areal turning radii in both branches. Their conserved j and ε follow from the same turning-point equations because the radial velocity vanishes there. The total-angle integrand differs through the radial response:

$$\frac{d\varphi}{d\xi} = \frac{\sqrt{\mathcal{H}(A(\xi))}}{\sqrt{1 - 2(3 + e\cos\xi)/P}}, \quad \mathcal{H} = A\zeta^2 . \quad (204)$$

The selected branch has $\mathcal{H} = 1$; the flat-radial control has $\mathcal{H} = A$. At $e = 0.2$, independent orbit integration and quadrature give:

Table 8. Alternative radial response and exact orbit checks

P	Radial response	Advance from quadrature rad	Advance from orbit integration rad
100	$\zeta^2 = 1$	0.1322932660	0.1322932660
100	$\zeta^2 = 1/A$	0.1974475160	0.1974475160
1000	$\zeta^2 = 1$	0.0126295320	0.0126295320
1000	$\zeta^2 = 1/A$	0.0189349966	0.0189349966

The agreement of two formulations within the flat-radial branch is exact, while the two radial branches give different accumulated azimuth. The difference is not removed by a consistent change of clock parameter. It is why the field-response selection must precede the orbital interpretation of $Q_\star$.

B.1 The weak field contribution of radial response

The leading difference can also be seen directly in the action. For this comparison only, write $\zeta^2 = A^{-\lambda}$, with $\lambda = 0$ for the control and $\lambda = 1$ for the selected response. Let $U_N = GM/r$ and $v^2 = v_r^2 + v_t^2$, with these coordinate velocities taken per T. Expansion gives

$$\frac{L_p}{m_p} = -c^2 + \frac{v^2}{2} + U_N + \frac{L_1}{c^2} + O(c^{-4}), \quad (205)$$

$$L_1 = \frac{v^4}{8} + \frac{U_N v^2}{2} + \lambda U_N v_r^2 + \frac{U_N^2}{2}. \quad (206)$$

To see the advance without inserting its coefficient, expand the exact general orbit equation at fixed $p_0 = j^2/(GM)$. With $\ell_g = GM/c^2$, the result is

$$w'' + w = \frac{1}{p_0} + \ell_g \left[(1+2\lambda)w^2 + \frac{2(1-\lambda)w}{p_0} + (1-\lambda)(w')^2 \right] + O\left(\frac{\ell_g^2}{p_0^3}\right). \quad (207)$$

Substitute $w_0 = (1 + e\cos\varphi)/p_0$ into the first-order terms. The resonant coefficient is $2(2+\lambda)e\ell_g/p_0^2$. The same resonance calculation used in Section 5.4 therefore gives

$$\Delta\varphi = \frac{2\pi(2+\lambda)GM}{pc^2} + O\left(\frac{G^2M^2}{p^2c^4}\right). \quad (208)$$

Table 9. The perihelion contribution of the radial response

Specified radial response	Leading coefficient	Mercury arcsec per century
$\zeta^2 = 1$	$4\pi GM/(pc^2)$	28.6538
$\zeta^2 = 1/A$	$6\pi GM/(pc^2)$	42.9807

The radial term in equation (206) vanishes instantaneously at periapsis because $v_r = 0$, yet contributes to the finite orbit as the planet approaches and leaves it. A calculation confined to the peak readout cannot capture that contribution. Splitting the total advance into time and spatial pieces depends on the chosen representation; the accumulated apsidal angle is the comparison quantity.

Responses differing only at higher weak-field order can preserve the leading 6π coefficient while changing exact orbits. Agreement with the leading Mercury number alone is therefore weaker than the source-family selection proved in Section 3.4.

Appendix C Receiving clock accumulation around perihelion

The normalized readout is greatest at perihelion, but its accumulated effect also depends on the time spent there. For a held clock and a moving clock, respectively, the leading rate changes are

$$\frac{Q_\star}{Q_0} - 1 = \frac{GM}{rc^2} + O(c^{-4}), \quad (209)$$

$$\frac{Q_{\mathsf{rec}}}{Q_0} - 1 = \frac{1}{c^2}\left(\frac{GM}{r} + \frac{v^2}{2}\right) + O(c^{-4}). \quad (210)$$

Use the leading Kepler orbit, which is sufficient for these first-order clock effects. If $\mathscr{E}$ denotes eccentric anomaly and $n = \sqrt{GM/a^3}$, then

$$r = a(1 - e\cos\mathscr{E}), \quad dT = \frac{1 - e\cos\mathscr{E}}{n}\, d\mathscr{E}. \quad (211)$$

The radius dependence cancels from the leading gravitational clock-deficit integral:

$$\Delta T_{\mathsf{h}} \equiv \int (1 - \alpha) dT \simeq \frac{GM}{ac^2 n}\, \Delta\mathscr{E}. \quad (212)$$

Thus the deficit is uniform per eccentric-anomaly increment at this order, although the instantaneous rate change is larger near perihelion. For a full orbit of period $P_T = 2\pi/n$, the gravitational and moving-clock deficits are

$$\Delta T_{\mathsf{h}} \simeq P_T \frac{GM}{ac^2}, \quad \Delta T_{\mathsf{moving}} \simeq \frac{3}{2} P_T \frac{GM}{ac^2}. \quad (213)$$

The second expression follows from $v^2 = GM(2/r - 1/a)$. With the Mercury inputs of Section 5.6, the self-consistent Kepler period is 87.96947 days. The resulting full-orbit deficits are 0.193806 seconds and 0.290710 seconds. These are calculated clock comparisons in the isolated solar model, not observations of a clock on Mercury.

A sum of held-clock increments along successive orbital locations is not the reading of one clock held at a fixed radius. The moving-clock quantity does follow the model orbit. Mercury's own gravitational field, other bodies and corrections beyond the stated approximation are not included in these numerical comparisons.

C.1 The finite interval around perihelion

Define a finite passage by true anomaly between $-30°$ and $+30°$. Its conversion to eccentric anomaly is

$$\tan\frac{\mathscr{E}}{2} = \sqrt{\frac{1-e}{1+e}} \tan\frac{\varphi}{2}. \quad (214)$$

The interval lasts 9.6021 days. The gravitational clock deficit is 26.4235 milliseconds; including orbital motion gives 42.2698 milliseconds. It contains 10.9153 percent of the orbital time and 13.6340 percent of the leading gravitational clock deficit.

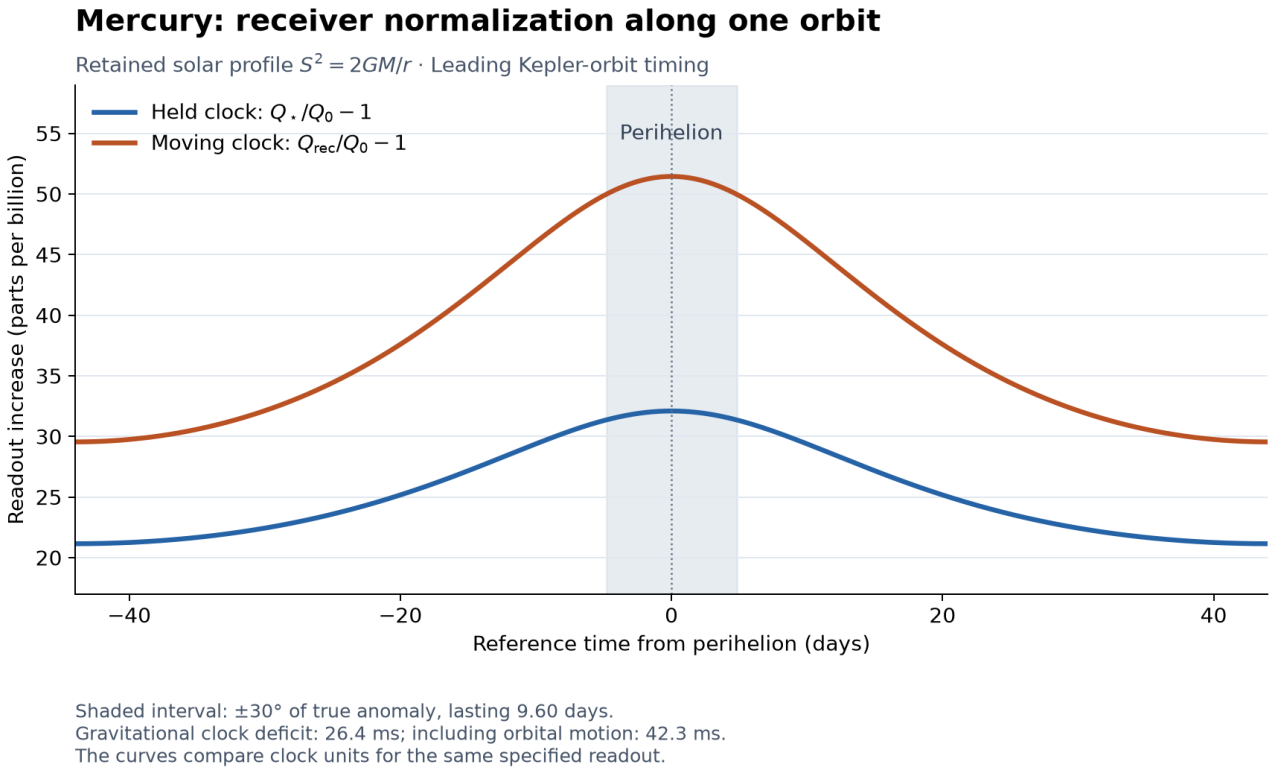


Figure 8. Receiving-clock normalization through one leading Mercury orbit in the retained solar field. The shaded region is the finite passage defined above. The held-clock increase is 32.10 parts per billion at perihelion and 21.15 at aphelion; the moving-clock increase is 51.45 and 29.55 parts per billion, respectively.

The time normalization conserves the same specified readout point by point:

$$Q_\star d\tau_{\mathsf{h}} = Q_0 dT, \quad \int Q_\star d\tau_{\mathsf{h}} = \int Q_0 dT\,. \quad (215)$$

This remains true when $Q_0(r)$ varies along the orbit. The larger rate per receiving-clock second does not provide an additional source quantity to add to the force. Its dynamical role is already carried by the normalized potential and time conversion derived in Section 5.1.

Appendix D The free fall representation

The main calculation used the capacity norm, physical work and source consistency. A coordinate representation provides an additional check. Before selecting $\mathscr{H}$, multiplication of the invariant by dT^2 gives

$$c^2 d\tau^2 = A c^2 dT^2 - \frac{\mathscr{H}}{A} dr^2 - r^2 d\Omega^2 . \quad (216)$$

Write $h = \sqrt{\mathscr{H}}$. For inward marginal-energy radial motion, $\varepsilon = 1$ and $j = 0$ give

$$\frac{dT}{d\tau} = \frac{1}{A}, \quad \frac{dr}{d\tau} = -\frac{S}{h} . \quad (217)$$

Define a time coordinate by

$$dt_f = dT + \frac{Sh}{Ac^2} dr . \quad (218)$$

The coefficient depends only on radius, so this is locally an integrable change of stationary synchronization. Along the stated free-fall trajectories, $dt_f/d\tau = 1/A - S^2/(Ac^2) = 1$. Substitution into equation (216) gives

$$c^2 d\tau^2 = c^2 dt_f^2 - (h\, dr + S\, dt_f)^2 - r^2 d\Omega^2 . \quad (219)$$

This identity holds for every positive $\mathscr{H}$ before selection. It identifies $d\chi = h\, dr = \alpha\, d\ell$ as the radial increment on these free-fall slices. The work-current defined in Section 3.3 is accordingly

$$\mathscr{J} = -\frac{1}{2}\frac{dS^2}{d\chi} = \frac{c^2}{2h}\frac{dA}{dr} . \quad (220)$$

After the source-family result sets $\mathscr{H} = 1$, equation (219) is the usual ingoing Painlevé–Gullstrand form of the selected exterior [14]. Its appearance is a consequence and coordinate crosscheck. No inference that S is a literal material outflow is required by this transformation.

The spatial volume element on this unit-energy free-fall slicing is $4\pi r^2 h\, dr$. It equals the spherical reference-shell volume after selection. That equality has not been used as an additional assumption to obtain $\mathscr{H} = 1$, and it is not asserted for all observer slicings.

D.1 The equivalent radial drift expression

The same selected exterior can be written with a signed radial drift Br. The following transformation records the equivalent representation used in Draft 48. Here it is applied after the spherical field has been determined.

$$c^2 d\tau^2 = c^2 dt_c^2 - (dr - B_r dt_c)^2 - r^2 d\Omega^2, \quad B_r^2 = S^2 . \quad (221)$$

Here tc is the time coordinate of the propagation representation, $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\varphi^2$, and Br is a signed radial drift variable. The radial displacement is dr − Br dtc, while the angular displacement is unchanged. This is the Painlevé–Gullstrand representation [3], used here for the already selected exterior. Its square contains the gravitational contribution, so adding a separate $S^2 dtc^2$ term would count it twice.

For dr = dΩ = 0, equation (221) gives $(d\tau/dtc)^2 = 1 - Br^2/c^2$. Matching the held-clock budget fixes $Br^2 = S^2$. Now remove the mixed radial–time term by defining:

$$dt_c = dT - \frac{B_r}{\alpha^2 c^2}\, dr\,. \quad (222)$$

Because the coefficient depends only on r, this transformation is integrable in the static exterior. Expansion and substitution give:

$$c^2 d\tau^2 = \alpha^2 c^2 dT^2 - \left[1 + \frac{B_r^2}{\alpha^2 c^2}\right] dr^2 - r^2 d\Omega^2\,. \quad (223)$$

The radial coefficient reduces algebraically:

$$1 + \frac{B_r^2}{\alpha^2 c^2} = 1 + \frac{u}{1-u} = \frac{1}{\alpha^2}\,. \quad (224)$$

Consequently:

$$c^2 d\tau^2 = \alpha^2 c^2 dT^2 - \frac{dr^2}{\alpha^2} - r^2 d\Omega^2\,. \quad (225)$$

The static spatial displacement is therefore $d\ell^2 = dr^2/\alpha^2 + r^2 d\Omega^2$. Substituting the spherical burden gives the Schwarzschild exterior interval. The metric records the clock and propagation relations obtained from the specified capacity, source and coupling equations.

Both signs of Br give the same static exterior interval because only Br^2 remains. In this calculation Br represents the propagation coupling, without establishing a literal material flow. It is different from Appendix F.1's gradient current $\mathcal{J}$: Br has units of speed and $\mathcal{J}$ of acceleration. We can eliminate Br from the final equations while retaining the spatial response it supplied.

Appendix E Galaxy inputs and uncertainty calculations

Galaxy	D Mpc	i degrees	Projected axis ratio	Radius status	Sources
NGC4051	18.00 ± 2.50	49.0 ± 3.0	0.880 ± 0.025	End	VS01,SV98
NGC5371	39.70 ± 9.92	53.0 ± 2.0	0.782 ± 0.012	Interior 0.50	Sa96,Be87
UGC07690	8.11 ± 2.43	41.0 ± 5.0	0.812 ± 0.035	Interior 0.57	Sw09,Sw02
NGC3949	18.00 ± 2.50	55.0 ± 2.0	0.669 ± 0.015	End	VS01,SV98
NGC4013	18.00 ± 2.50	89.0 ± 1.0	0.405 ± 0.010	Interior 0.43	VS01,SV98
UGC06818	18.00 ± 2.50	75.0 ± 3.0	0.467 ± 0.048	Interior 0.75	VS01,SV98
NGC2683	9.81 ± 0.49	80.0 ± 5.0	0.351 ± 0.018	Interior 0.33	Sa96,Br92
NGC3198	13.80 ± 1.40	73.0 ± 3.0	0.398 ± 0.014	Interior 0.36	Da06,Be91,Be87
NGC3992	23.70 ± 2.30	56.0 ± 2.0	0.594 ± 0.017	Interior 0.60	VS01,SV98
UGC08837	7.21 ± 0.36	80.0 ± 5.0	0.277 ± 0.003	End	Sw09,Sw02
NGC5585	7.06 ± 2.12	51.0 ± 2.0	0.650 ± 0.028	Interior 0.56	Bl99,Sa96,Co91
NGC4010	18.00 ± 2.50	89.0 ± 1.0	0.198 ± 0.017	End	VS01,SV98
NGC4565	9.7 fixed	86.6 fixed	0.205 ± 0.017	End	Yim14,Salo15
NGC4183	18.00 ± 2.50	82.0 ± 2.0	0.239 ± 0.016	Interior 0.54	VS01,SV98
NGC5907	17.30 ± 0.90	88.0 ± 2.0	0.197 ± 0.010	Interior 0.55	Sa96,SA87
NGC0100	13.50 ± 4.05	89.0 ± 1.0	0.185 ± 0.028	Interior 0.93	dB02,Rh96
NGC4100	18.00 ± 2.50	73.0 ± 2.0	0.341 ± 0.010	Interior 0.61	VS01,SV98
NGC5033	15.70 ± 4.70	66.0 ± 1.0	0.431 ± 0.019	Interior 0.54	Sa96,Be87
UGC07089	18.00 ± 2.50	80.0 ± 3.0	0.228 ± 0.004	End	VS01,SV98
NGC1090	37.00 ± 9.25	64.0 ± 3.0	0.458 ± 0.042	Interior 0.79	Ge04

Table 10. Central inputs and comparison-radius status. Distances, inclinations and their listed errors are from the adopted catalogue except NGC 4565. Source codes follow SPARC Table 1 [5]; Yim14 and Salo15 refer to [7] and [8]. The projected axis-ratio errors are formal image-fit errors. “Interior” and “End” refer to the adopted rotation dataset; they do not define a unique physical galaxy edge.

Galaxy	R range kpc	Mbar range 10^{10} M⊙	Reported ratio range	Adds M/L sensitivity
NGC4051	10.50–13.88	3.782–6.672	0.834–1.248	0.750–1.390
NGC5371	17.34–28.90	10.361–29.058	0.844–1.178	0.760–1.311
UGC07690	1.65–3.07	0.046–0.162	0.741–1.675	0.706–1.772
NGC3949	6.09–8.05	1.733–3.068	0.848–1.456	0.775–1.601
NGC4013	11.60–15.34	3.439–6.066	0.917–1.149	0.826–1.278
UGC06818	4.51–5.97	0.163–0.293	0.768–1.169	0.740–1.223
NGC2683	10.96–12.12	3.885–4.770	0.988–1.183	0.885–1.322
NGC3198	14.44–17.70	2.704–4.089	0.965–1.199	0.907–1.284
NGC3992	24.85–30.19	11.017–16.363	0.850–1.163	0.774–1.282
UGC08837	3.99–4.41	0.060–0.075	0.882–1.150	0.848–1.205

Galaxy	R range kpc	Mbar range 10^{10} M⊙	Reported ratio range	Adds M/L sensitivity
NGC5585	4.31–8.01	0.181–0.630	0.760–1.377	0.728–1.447
NGC4010	9.02–11.92	0.910–1.616	0.948–1.199	0.877–1.303
NGC4565	Unreported	Unreported	0.868–1.081	0.786–1.197
NGC4183	9.77–12.93	0.742–1.317	0.887–1.197	0.837–1.278
NGC5907	26.24–29.12	10.366–12.838	1.027–1.127	0.944–1.233
NGC0100	6.23–11.57	0.207–0.726	0.709–1.205	0.681–1.263
NGC4100	12.00–15.88	2.496–4.408	0.907–1.200	0.822–1.330
NGC5033	16.82–31.20	3.687–12.756	0.774–1.202	0.708–1.321
UGC07089	7.89–10.43	0.249–0.448	0.788–1.184	0.744–1.263
NGC1090	17.92–29.86	2.661–7.516	0.807–1.294	0.742–1.415

Table 11. Distance-coupled radius ranges and mass ranges, plus predicted-to-observed speed-ratio envelopes. Mass ranges in this table use reported distance and luminosity variation at fixed adopted stellar M/L. The final column adds the separate ±0.1 dex stellar M/L sensitivity. Each range permits simultaneous variation of its listed inputs; it is not a statistical confidence interval. NGC 4565 has no adopted distance or luminosity uncertainty here, so its mass and radius ranges are unreported, not zero-uncertainty measurements.

The envelope evaluates a 601 by 601 grid in projected axis ratio and inclination, retaining only physical oblate deprojections. For each geometry it rescales the observed velocity using that same inclination. Distance, luminosity and quoted velocity errors enter through their monotonic endpoints. No mass or radius is changed in the central Table 3 calculation. The source hashes, central rows, envelope assumptions and reproduction code accompany the manuscript. The ±0.1 dex exercise rescales all stellar components together; it is a model sensitivity, not an additional measured error for each galaxy. Common group-distance and mass-to-light systematics are not independent trials across the sample.

The envelopes omit unreported HI-flux and decomposition systematics. NGC 4565 retains fixed distance and inclination because their uncertainties were not recovered. The separate stellar mass-to-light range is a model sensitivity. Agreement within an individual envelope does not establish a simultaneous solution for all galaxies under one shared set of systematic parameters.

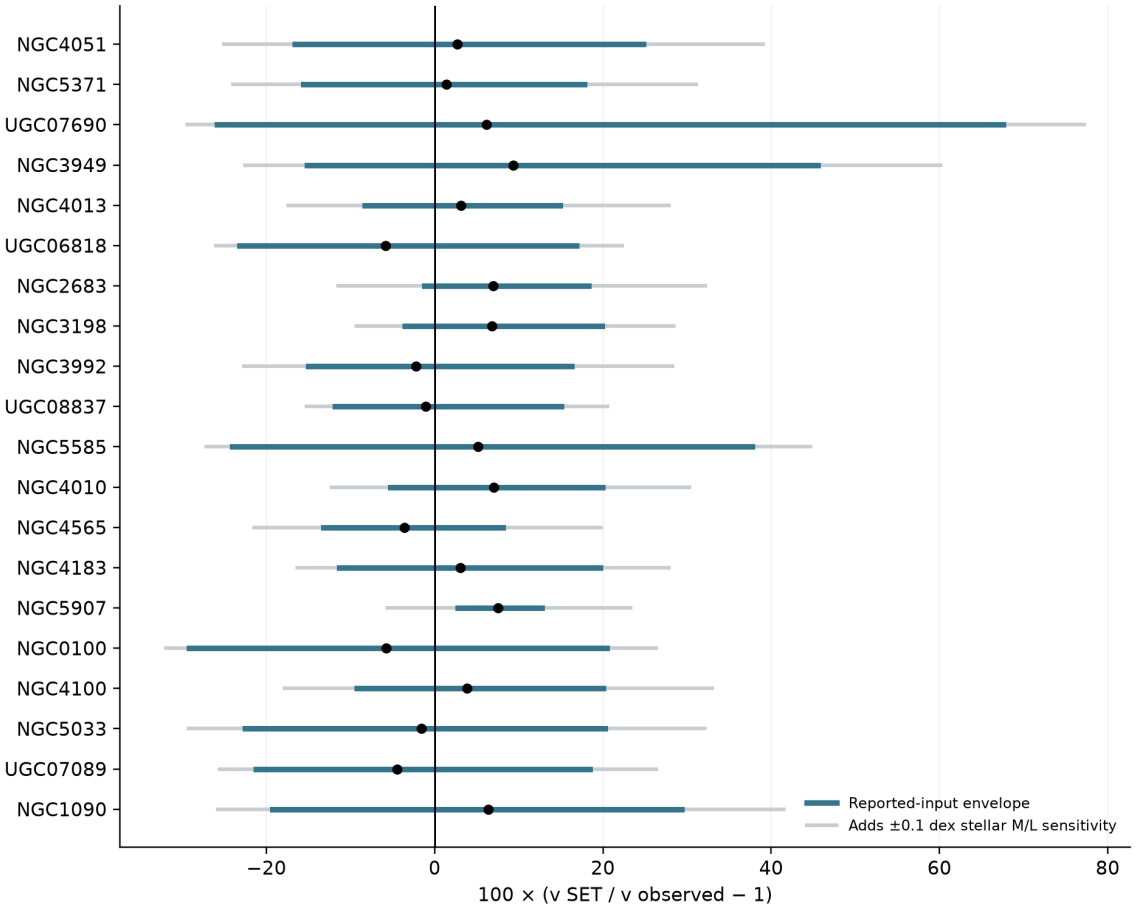


Figure 9. Central residuals and coupled-input envelopes for the selected examples. Blue lines use the reported-input ranges included in this calculation. Gray lines additionally vary the stellar mass-to-light normalization by ±0.1 dex. The two envelopes answer different questions. NGC 4565 retains fixed distance and inclination here. Unknown systematics and correlations between different galaxies are not assigned a probability model.

The projected stellar-aperture checks retain global HI and therefore are not complete enclosed-baryon reconstructions. For UGC 8837, the stellar correction changes the central discrepancy from −1.10% to approximately −1.55%; its close agreement survives that particular check. NGC 4565 has no equivalent enclosed-inventory check in this analysis. A comparison at an interior radius does not become a fully matched mass aperture merely because its central speed agrees.

Figure 5 uses a separate probabilistic input exercise, distinct from the endpoint envelopes in Table 11 and Figure 9. Its blue bars are the marginal 2.5th and 97.5th percentiles of 65,536 joint draws, with stellar M/L fixed at the adopted values. Quoted errors set Gaussian widths for projected axis ratio, inclination, distance, luminosity and speed; positive quantities and physical oblate deprojections are retained. Distance rescales mass and radius together, and the same inclination draw changes both the reconstructed area and the deprojected speed. The two plotted axes therefore have correlated errors.

The retained distance model includes a common 0.9 Mpc term for Ursa Major members and a common 7% term for Hubble-distance objects, with the remaining quoted variance assigned individually. These are assumptions of the sensitivity model. Gas flux and other unreported systematics are held fixed; NGC 4565 also retains fixed distance, inclination and luminosity. The archived draw code records the distributions and seed. These conditional percentiles do not include a complete mass-model covariance or selection adjustment, and the orange central predictions have no comparable full uncertainty distribution supplied here.

Appendix F Stationary source response and energy balance

F.1 Spherical charge and gradient flux

Section 3 defines the source strength $\mu Q = Q\Omega^2/(24\pi V\Omega) = GM$. For a Gauss-law representation, set $\Psi = S^2/2$ and $\mathcal{J} = -\nabla\Psi$. This outward diagnostic current has units m/s^2 and its surface integral m^3/s^2.

For the isotropic stationary branch, impose $\nabla\cdot\mathcal{J} = 4\pi G\rho$. Integration over a domain containing mass M gives $\oint \mathcal{J}\cdot dA = 4\pi GM = Q\Omega^2/(6V\Omega)$. In its spherical source-free exterior, symmetry therefore gives $\mathcal{J}r = GM/r^2$. Substituting $\mathcal{J}r = -d\Psi/dr = -\frac{1}{2}dS^2/dr$ immediately yields $-(r^2/2)dS^2/dr = GM$, equation (41). The boundary condition $\Psi \to 0$ at infinity gives $\Psi = GM/r$ and $S^2 = 2GM/r$.

The source identity fixes the integrated normalization, while $\mathcal{J} = -\nabla\Psi$ specifies the isotropic response. Here $\mathcal{J}$ is shorthand for the gradient and can be eliminated. The spherical areal radius is defined by $4\pi r^2$, no auxiliary radius calibration enters this calculation.

In the conditional collective extension, the corresponding representation is $\mathcal{J} = -C\nabla\Psi$. The tensor C belongs to that proposed response law, not to the source identity or to the invariant's remaining-capacity fraction. Appendix F states that stationary formulation. The stationary source equations determine the burden field for the specified response. The spherical clock and propagation relations follow from the physical-current and source-consistency construction in Section 3.

Earlier manuscripts developed this source and action language [1, 2]. The research supplement retains the previous auxiliary radial calibration for traceability. The source and current relations used here are given explicitly.

F.2 A stationary collective field

I now state a stationary source equation for a prescribed spatial response. Use Euclidean spatial derivatives in the weak field and a prescribed dimensionless symmetric positive tensor C:

$$-\nabla\cdot(C\,\nabla\Psi) = 4\pi G\rho_{\mathrm{bar}},\quad \Psi = \frac{S^2}{2},\quad \boldsymbol{a} = \nabla\Psi\,. \quad (226)$$

Equivalently, the source equation is $-\frac{1}{2}\nabla\cdot(C\nabla S^2) = 4\pi G\rho\mathrm{bar}$. Let fixed sources have normalized density profiles w_a, with $\int w_a\, d^3x = 1$, and define $\mu_a = Q_a^2/(24\pi V_a) = GM_a$. Its right side can then be written $4\pi\Sigma\mu_a w_a$. Moving sources changes their locations and thus the density; it need not change their individual M_a, V_a or Q_a. Splitting the same density into more numerical cells leaves this source unchanged.

The same source normalization can be written without abbreviating Q:

$$-\frac{1}{2}\nabla\cdot(C\,\nabla S^2) = \sum\nolimits_a \frac{[4\pi\sqrt{2GM_aR_a^3}]^2}{6(4/3)\pi R_a^3} w_a(\boldsymbol{x}) = 4\pi G\rho_{\mathrm{bar}}(\boldsymbol{x})\,. \quad (227)$$

For the specified constant response and isolated boundary condition, equation (228) gives $S^2 = 2\Psi$. Substituting that solution into equation (1), before simplifying the source coefficient, produces equation (168).

Define $\mathcal{J} = -C\nabla\Psi$. Any complete enclosing surface then has $\oint \mathcal{J}\cdot dA = 4\pi GM$. The source normalization is fixed while C determines how that gradient flux relates to the force $\nabla\Psi$. C contains response coefficients.

For constant positive C and an isolated boundary condition $\Psi \to 0$ at infinity, the solution is explicit. Write d = x − x′, C^{-1} for the matrix inverse and det C for its determinant:

$$\Psi(\boldsymbol{x}) = \frac{G}{\sqrt{\det C}} \int \frac{\rho_{\mathrm{bar}}(\boldsymbol{x}')}{\sqrt{\boldsymbol{d}^T C^{-1} \boldsymbol{d}}}\, d^3x' . \quad (228)$$

The orientation at an enclosing boundary follows from this same solution. For a positive point source M_a at x_a, define $d_a = x - x_a$. Differentiating equation (228), with C constant and positive-definite, and using $\mathcal{J} = -C\nabla\Psi$ gives:

$$\mathcal{J}_a(\boldsymbol{x}) = \frac{GM_a}{\sqrt{\det C}} \frac{\boldsymbol{d}_a}{(\boldsymbol{d}_a^T C^{-1} \boldsymbol{d}_a)^{3/2}} . \quad (229)$$

Let n be the outward unit normal at x on a convex surface enclosing the sources, with x distinct from their locations. Geometry gives $d_a \cdot n \geq 0$. Since the coefficient multiplying d_a is positive:

$$\boldsymbol{d}_a \cdot \boldsymbol{n} \geq 0 \;\Rightarrow\; \mathcal{J}_a \cdot \boldsymbol{n} \geq 0, \;\; \mathcal{J} \cdot \boldsymbol{n} = \sum\nolimits_a \mathcal{J}_a \cdot \boldsymbol{n} \geq 0. \quad (230)$$

The outward-normal contribution is strictly positive for a source strictly inside the surface. Summing positive source contributions preserves the sign; integrating over a nonnegative continuous density gives the same result. $\mathcal{J}$ retains its defined gradient-flux meaning. This orientation property does not establish QΣ = Qref; that boundary-loading normalization remains the collective prescription in Section 7.3.

The coordinate change $y = C^{-1/2}x$ converts the operator to an ordinary Laplacian and transforms the source volume. With C = I, equation (228) becomes the positive Newtonian potential. For specified constant C, equation (228) determines the stationary field of an arbitrary specified density.

Multiplying Ψ by a constant and dividing C by that constant preserves $\mathcal{J}$ and its source equation.

F.3 Work when the response changes

Energy provides a consistency test of the proposed response. Use a weak-field quasistatic family of equation (226), with finite smooth sources, Ψ fixed to zero at infinity and a symmetric positive C(x,T) prescribed independently of the source variations. Let matter obey mass continuity and the force $a = \nabla\Psi$. For the gravity-only case define:

$$E_{\mathrm{kin}} = \frac{1}{2} \int \rho\, |\boldsymbol{v}|^2 \, d^3x, \quad W = -\frac{1}{2} \int \rho\Psi \, d^3x . \quad (231)$$

The factor ½ in the binding energy avoids counting each source interaction twice. Multiplying the source equation by Ψ and integrating by parts, with vanishing boundary terms, gives:

$$W = -\frac{1}{8\pi G} \int \nabla\Psi \cdot C\, \nabla\Psi \, d^3x . \quad (232)$$

These expressions refer to a finite-energy source distribution. Unregularized point-source self-energies cannot be inserted into them. If pressure, internal energy, radiation or external matter forces are present, their corresponding energy and work terms must also be included.

The same source equation and force follow from a compatible quasistatic Lagrangian:

$$L_{\mathrm{qs}} = E_{\mathrm{kin}} + \int \rho\Psi \, d^3x - \frac{1}{8\pi G} \int \nabla\Psi \cdot C\, \nabla\Psi \, d^3x . \quad (233)$$

C is held fixed when varying Ψ and the source positions in this expression. To show the energy balance, define:

$$P_{\text{response}} = \frac{1}{8\pi G}\int \nabla\Psi \cdot \frac{\partial C}{\partial T}\nabla\Psi \, d^3x \,. \quad (234)$$

Differentiate the source equation and use its self-adjoint spatial operator. Integration by parts gives $\int\rho(\partial\Psi/\partial T)\, d^3x = \int\Psi(\partial\rho/\partial T)\, d^3x - 2P_{\text{response}}$. The partial derivatives hold spatial coordinates fixed. Differentiate W in equation (231) and substitute this identity:

$$\frac{dW}{dT} = -\int \Psi \frac{\partial \rho}{\partial T} d^3x + P_{\text{response}} \,. \quad (235)$$

Mass continuity, $\partial\rho/\partial T = -\nabla\cdot(\rho v)$, and the gravitational force give $dE_{kin}/dT = \int\rho v\cdot\nabla\Psi\, d^3x = \int\Psi(\partial\rho/\partial T)\, d^3x$ under the same boundary assumptions. Therefore:

$$\frac{d(E_{\text{kin}} + W)}{dT} = \frac{1}{8\pi G}\int \nabla\Psi \cdot \frac{\partial C}{\partial T}\nabla\Psi \, d^3x \,. \quad (236)$$

The right side has units of power. Fixed C conserves Ekin + W. If C changes externally, it supplies or removes work through the stated term. As a sign check, take a uniform scalar C = kI at fixed density: $\Psi = \Psi_0/k$ and $W = W_0/k$. Since $W_0 < 0$, increasing k makes W less negative, as equation (236) requires.

The interpretation changes if C is determined by the sources themselves. At fixed boundary conditions the full variation is:

$$\delta W = -\int \Psi\, \delta\rho \, d^3x + \frac{1}{8\pi G}\int \nabla\Psi \cdot \delta C \, \nabla\Psi \, d^3x \,. \quad (237)$$

If a source coordinate Xa changes C, this second term contributes an additional generalized force:

$$F_{a,\text{response}} = -\frac{1}{8\pi G}\int \nabla\Psi \cdot \frac{\partial C}{\partial X_a}\nabla\Psi \, d^3x \,. \quad (238)$$

Its work cancels the part of Presponse caused by that internal coordinate change. Updating C from a rearranging source distribution while retaining only the force ∇Ψ would omit that interaction. Alternatively, C can be assigned independent dynamics and its own energy, but those equations must then be supplied. Equation (236) alone is not a closed evolution law for a galaxy.

Appendix G A finite flat speed profile and its exterior

To illustrate the relation between potential depth and orbital speed, prescribe a weak-field circular speed v_0 from an inner radius Rin > 0 to a transition radius Rt. Outside Rt, prescribe $v_{orb}^2(R) = v_0^2 R_t/R$. Both the speed and the force are continuous at Rt, and the inverse-radius exterior allows S^2 to approach zero at infinity.

Integrating equation (89) first in the exterior gives:

$$S^2(R) = \frac{2v_0^2 R_t}{R}, \quad R \geq R_t . \quad (239)$$

Inside the flat-speed interval, split the same integral at Rt:

$$S^2(R) = 2\int_R^{R_t} \frac{v_0^2}{r}\, dr + 2v_0^2 . \quad (240)$$

$$S^2(R) = 2v_0^2\left[1 + \ln\left(\frac{R_t}{R}\right)\right], \quad R_{in} \leq R \leq R_t . \quad (241)$$

Differentiation gives $-(R/2)dS^2/dR = v_0^2$ throughout that interval. At Rt, both sides give $S^2 = 2v_0^2$ and $dS^2/dR = -2v_0^2/R_t$. Thus the boundary conversion $S/\sqrt{2} = v_0$ is consistent at the transition, even though it does not hold at every interior point. For example, at R = Rt/2, $S/\sqrt{2} = v_0\sqrt{(1 + \ln 2)} \approx 1.3012 v_0$ while the circular speed remains v_0.

Below Rin the model needs an inner solution. Its exterior gravitational parameter is $\mu_{ext} = v_0^2 R_t$, or $\mu_{ext} = (4/f)^2 GM_{bar}$ under the boundary calibration at Rt. Thus a choice $f \neq 4$ retains an enhanced exterior monopole. This construction joins a flat-speed segment to the edge conversion; the source law would still have to determine Rt, f and the profile.

Appendix H Quadratic gradient normalization

The current conversion also supplies a check for any expression quadratic in the burden gradient. Squaring equation (70) gives

$$\left(\frac{dS^2}{d\chi}\right)^2 = \alpha^8\left(\frac{ds_\star^2}{d\chi}\right)^2 . \quad (242)$$

Thus, for any specified weight $W(\chi)$ and the same integration measure,

$$\mathcal{I}_S = \int W(\chi)\left(\frac{dS^2}{d\chi}\right)^2 d\chi, \quad (243)$$

$$\mathcal{I}_S = \int W(\chi)\alpha^8\left(\frac{ds_\star^2}{d\chi}\right)^2 d\chi . \quad (244)$$

If W depends on the field, its dependence must also be expressed in the new variable. The equations above only convert an already specified expression. Its units, physical role, integration domain and weight remain to be supplied by that expression; it is not automatically a field energy. Omitting α^8 would generally give a different functional rather than a different notation for the same one.

The finite-interval counterpart follows directly from equation (69):

$$\mathcal{J}_{12}^2 = \alpha_1^4\alpha_2^4\left(\frac{s_{\star 1}^2 - s_{\star 2}^2}{2\Delta\chi}\right)^2 . \quad (245)$$

These identities are useful when comparing a proposed field action written in capacity burden with one written in receiving-clock units. They prevent a change of normalization from silently changing the dynamics. They do not select a field-energy functional, and none is added to the spherical field construction here.

The main derivation needs the first-gradient conversion in Sections 4.2 and 4.3 and the transformed motion equation in Section 5.1. Those checks, together with this quadratic identity, keep the physical work-current, the auxiliary orbital force and a possible quadratic field expression distinct. The orbit and Mercury benchmark remain those derived in Sections 5.3 and 5.6.

Appendix I Numerical and algebraic verification

The coordinate transformation was checked for both signs of the radial drift at six radii and multiple non-null displacement increments. The largest scaled residual was 1.6e-15. The analytic oblate surface coefficient was also compared with a direct numerical surface integral at six axis ratios; the maximum coefficient difference was 4.9e-15.

A separate trajectory integration advances the spatial motion law, rather than the energy first integral, in units c = Rs = m = 1. For p = 8Rs and eccentricity 0.3, fourth-order Runge–Kutta steps of 0.4 and 0.2 over proper duration 2400 both give nine perihelion passages. Maximum relative energy drift falls from 1.57e-11 to 8.8e-13 as the step is halved. The finer trajectory gives mean perihelion advance 1.6698605690 radians, compared with 1.6698605710 from the independent anomaly integral.

Mercury's weak-field integral uses a cancellation-resistant form of $(1 - z)^{-1/2} - 1$, evaluated by Gaussian quadrature. It gives 42.98048057 arcsec per Julian century. The Earth normalization and radial signal delay were recalculated from the stated inputs. The checks in this appendix verify algebra and implementation; observational comparisons are reported separately.

The source-to-boundary identities were checked over multiple masses, radii and area coefficients, including spherical and thin-area limits. Their largest relative numerical discrepancy was 6.7×10^{-16}. The exact spherical escape condition $\alpha\,\gamma_{escape} = 1$ was checked at five occupancies, with a largest energy-normalization discrepancy of 1.2×10^{-15}. Numerical differentiation of Appendix G's matched profile recovered its stipulated circular-speed function to within 8×10^{-11} in normalized squared-speed units.

The response-work identity was independently checked with finite source distributions and symmetric positive tensors, including changing eigenvalues, rotating axes and spatially varying response. A three-dimensional finite-element differentiation check had maximum normalized disagreement below 8.2×10^{-11}. A separate evolving-sphere calculation checked integrated mechanical-energy change against response work for constant, increasing, decreasing and cyclic prescribed coefficients. Halving the integration step reduced the fourth-order integration error by approximately sixteen. The sign and normalization of the work term passed those tests. Their assumptions, inputs and scripts accompany the research record.

For Mercury, the archived JPL-based example uses $GM_{\odot} = 1.3271244004127942\times10^{20}\ m^3/s^2$, $a = 5.790922654\times10^{10}$ m, e = 0.20563593 and period 87.96946593 days [9]. The result is about 0.1035 arcsec per orbit, or 42.98 arcsec per Julian century. Treating these inputs as exact turning-point parameters, the exact integral differs from the leading term by approximately 5.16×10^{-6} arcsec per century. That is a mathematical comparison, without converting fitted ephemeris elements to second post-Newtonian accuracy. The model uses an isolated spherical Sun and a test probe. Planetary and other contributions must be added when comparing with the total observed precession [4].

Appendix J Cosmological distance and domain checks

The grouped brightness result concerns the redshift dependence of relative flux. The checks below retain the same source inputs and distinguish calculated quantities from the observations and inference procedures used to assess them.

The absolute rate and nearby distances

The source and clock prescription gives $H_0 = 30.54\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. The cited megamaser analysis infers $73.9 \pm 3.0\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ from geometric distances and galaxy redshifts [19]. This is not a ΛCDM prediction. Its comparison with SET depends on how the model connects its calculated rate to those distances and redshifts.

Under the propagation relation retained in Section 11.1, the leading small-redshift distance is $D_L \simeq cz/H_0$. The fixed inputs therefore predict a nearby distance–redshift slope that does not reproduce the megamaser inference under that mapping. The relative supernova contrasts cancel the common absolute distance scale and do not test this normalization.

The BAO geometry comparison

The separate comparison with six released BAO geometry quantities gives $\chi^2 = 48.70$ [20]. This check uses the dimensionless ratio of transverse to radial distance, for which the common acoustic-ruler scale cancels. The released constraints still depend on the survey analysis and its standard-ruler interpretation. Under the retained flat propagation coupling, the statistic indicates poor agreement for this realization. It tests additional distance geometry rather than equality with ΛCDM parameters, and is separate from the supernova brightness statistics.

The domain of the retained history

The physical domain of this history ends at $z = 2.7244$. This is a limit of the retained source and front evolution, not an observed edge of the universe. The supernova groups reported here lie within that domain. Extrapolation beyond it is not supplied by this branch.

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